RAVEL: A Spatial Coding Harness for Reasoning about Change and Preservation

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Abstract

A physical action can change one relation while preserving another involving the same moving object. We study selective spatial relation reasoning: jointly establishing relation endpoints and distinguishing change from preservation within and outside an action's participation scope. Our dataset contains 12,000 indoor questions, including a 4,000-question test set spanning observed updates, unseen consequences, constrained edits, and consecutive actions. We introduce Relation-Aware Validity and Evidence Linking (RAVEL), a training-free harness that models scenes, actions, relation dependencies, and evidence as a structured spatial representation in code. The harness executes its semantics to update valid attributes, withdraw invalid premises, and combine supported computations with a frozen vision-language model's judgments into joint answers. The complete configuration improves whole-question accuracy by 13.35–19.80 percentage points across five backends, using additional initial geometry and statewise calls. Matching raw information and call schedules on 500 questions retains gains of 18.20 points for Qwen3-VL-4B and 2.20 points for Astra. Controlled interventions locate Qwen's largest increment in executable organization, with smaller, backend-dependent binding effects. Joint relation scoring shows improvements in both change and preservation, while trajectory and provenance analyses expose unresolved visual judgments and retained initial mistakes. These results connect the validity of relational premises to the quality of complete answers under physical actions.

1. Introduction

Understanding an action requires reasoning about the relations it changes together with those it preserves. In Figure 1, releasing A makes it collide with B and causes B to move. A–B height and contact change, B remains on the floor, and fixed C–D direction persists. The same motion of B thus accompanies both change and preservation. A correct account must establish each relation's endpoints and distinguish preservation after participation from preservation outside the interaction's scope. We call this *selective spatial relation reasoning*.

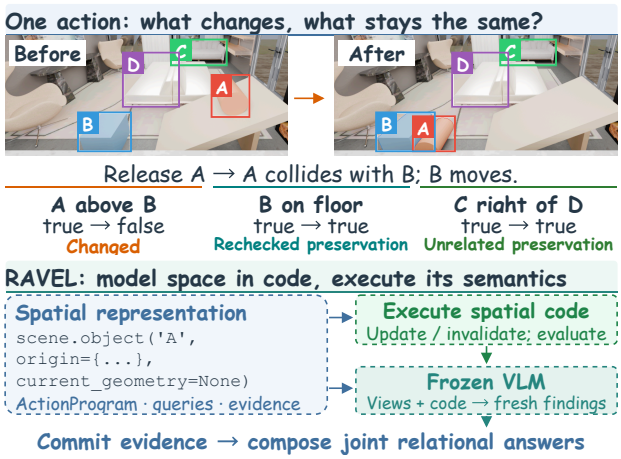


Figure 1. **One action, different relational consequences.** B moves yet remains supported; fixed C–D direction also persists. RAVEL models the scene, actions, dependencies, and evidence in code (fields abridged). Its harness executes attribute and query semantics, obtains unresolved findings from a frozen VLM, and composes joint answers. Both Astra methods solve this recorded example. Crops are identical; image access varies by task.

Existing work tests spatial estimation, dynamic memory, and physical prediction [2–4, 8, 12, 13]. We bring relation truth and action participation into a joint evaluation: detecting a collision is insufficient if the same answer loses a preserved support relation. Our benchmark spans observed transitions, unseen outcomes, multi-relation updates, constrained edit selection, and consecutive actions, with 12,000 questions and a 4,000-question test set. Preservation includes false relations: an edit that achieves its goal but violates a required unchanged relation is inadmissible.

The central inference problem is deciding which premises still establish a relation after an action. A prescribed translation can update a known center and support several direction or proximity judgments. Free motion invalidates that center as evidence of the final pose, even though the declaration remains in history. Meanwhile, fixed references can retain valid geometry and prior observations.

Participation alone cannot resolve these judgments: a moving object can preserve support while changing its distance to another object. The solver needs current endpoint evidence before interpreting change and preservation.

We introduce Relation-Aware Validity and Evidence Linking (RAVEL), a harness built around a structured spatial representation in code. Scene declarations encode object origins, reference frames, and valid attributes; action programs define how these attributes evolve; queries declare their dependencies and associated evidence. The harness interprets this representation to propagate or invalidate premises, compute supported relations, and ask a frozen vision-language model for unresolved findings. It commits these findings and composes joint answers about change, preservation, or admissible edits. The representation supplies both model context and harness execution semantics.

Across five backends, the full configuration improves whole-question accuracy by 13.35–19.80 points. Since it adds public initial geometry and state-wise calls, we match both factors on 500 questions: Qwen improves from 6.20% to 24.40%, and Astra from 78.00% to 80.20%. Binding and geometry interventions distinguish the organization bundle from individual enforcement rules. Relation-level gains extend to both change and preservation for Qwen; complete trajectories remain difficult, and provenance reveals both repeated fresh errors and reused initial mistakes.

Our contributions are a benchmark that jointly tests relation truth and action participation; a spatial coding harness linking executable declarations to evidence and joint decisions; and an evaluation combining five-backend comparisons, matched interventions, and relation-level diagnosis.

2. Related Work

Spatial and physical reasoning. SpatialVLM [3] develops large-scale spatial question answering; DSI-Bench [13] decouples observer and object motion; SpaMEM [8] studies action-conditioned spatial belief evolution. CLEVRER [12], Physion [2], and PhysBench [4] evaluate complementary aspects of physical events and prediction (Tab. 1). Our benchmark couples relation endpoints with execution participation, separating two forms of preservation across five task interfaces. HSSD [7] and Infinigen Indoors [9] supply the indoor scene sources.

Executable reasoning and harnesses. VISPROG [5] and ViperGPT [11] compose executable visual modules. CROSS [10] uses typed geometric operators for video spatial reasoning. VISTA [6] retains visual history and lets a model actively revisit images through a visual harness. RAVEL encodes scenes, actions, and relational evidence in a structured spatial representation with explicit execution semantics. Its harness interprets attribute validity and query dependencies, then composes computed and model-provided findings under a fixed call schedule. The representation is compiled from public declarations; the model supplies observations rather than generating programs or selecting tools.

Table 1. **Evaluation targets and inference mechanisms.** Top: benchmark scope. Bottom: how evidence enters executable reasoning. Entries summarize documented mechanisms.

A. Benchmarks	**Evaluation target**
SpatialVLM [3]	Qualitative and metric spatial questions
DSI-Bench [13]	Observer and object motion
SpaMEM [8]	Action-conditioned spatial memory
CLEVRER [12]	Descriptive, explanatory, predictive, and counterfactual collision reasoning
Physion [2]	Prediction across physical phenomena
PhysBench [4]	Physical properties, relations, scenes, dynamics
Our benchmark	Joint endpoints and participation; change, two preservation types, edits, consecutive updates
B. Harnesses	**Execution and evidence mechanism**
VISPROG [5]	Generated programs compose visual modules
ViperGPT [11]	Python programs combine vision APIs
CROSS [10]	Typed geometric operators and spatial skills
VISTA [6]	Active image revisiting; raw visual history
Direct VLM	One call; final-answer schema
RAVEL harness	Spatial code; action-dependent attribute validity; typed evidence; joint composition

3. Benchmark: Selective Relation Reasoning

3.1. Relation Truth and Participation

An episode pairs states $s_0, \ldots, s_{L-1}$ with actions $a_0, \ldots, a_{L-2}$ and queries $\mathcal{Q}$. Objects $\{A, B, C, D\}$ have persistent red/blue/green/purple identities and include controlled, free, and fixed participants. Each state has two views, subject to task permissions. Relation values $v_{t,q} \in \{\text{true}, \text{false}\} \cup \{\bot\}$ are true, false, or unknown at a threshold boundary or under insufficient evidence.

The seven predicates in Figure 2 use center, surface, interior, or joint geometry. Distances use meters in the fixed initial-view-1 ground frame, with vertical z. Direction and proximity use centers; `on_surface` denotes bottom contact or proximity to a named upward surface, not force equilibrium (Appendix B).

Let $\mathcal{D}(q)$ contain the query's named objects and $\mathcal{A}_t$ the directly controlled or secondarily moving participants. An update combines endpoint truth and participation:

$$u_{t,q} = \begin{cases} \textsf{unknown}, & \bot \in \{v_{t,q}, v_{t+1,q}\}, \\ \textsf{changed}, & v_{t,q} \neq v_{t+1,q}, \\ \textsf{rechecked_preserved}, & \mathcal{D}(q) \cap \mathcal{A}_t \neq \varnothing, \\ \textsf{unrelated_preserved}, & \text{otherwise.} \end{cases} \tag{1}$$

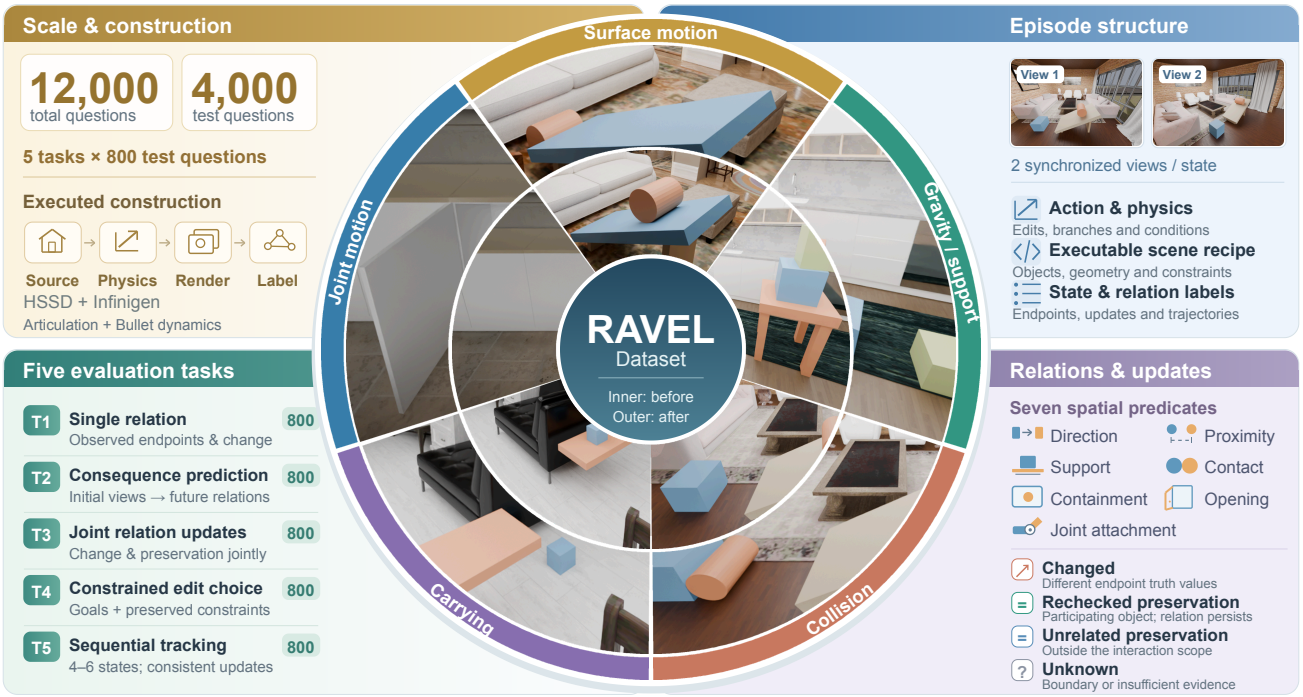


Figure 2. **RAVEL dataset overview.** The dataset contains 12,000 questions, including 4,000 test questions with 800 per task. Paired scene views illustrate five physical categories (inner: before; outer: after). Construction, episode structure, task interfaces, and relation labels define the benchmark. Image permissions depend on the task (Tab. 6).

The last two cases require known, equal values, including false-to-false preservation. Thus B–floor can remain true while B moves, making it rechecked; fixed C–D direction is unrelated. Gold participation describes recorded execution scope, rather than counterfactual causal effects.

3.2. Tasks, Construction, and Evidence Access

The dataset contains 12,000 questions from HSSD and Infinigen scenes; the test set has 4,000 questions, 800 per task, from 100 source groups. Tasks cover joint motion, gravity/-support, surface motion, collision, and carrying. Groups share layouts or assets and define statistical dependence.

T1/T3 score observed endpoints and three-/four-way updates for single/multiple relations; T1 merges preservation types. T2 predicts future relations from initial views and actions. T4 selects all independent edits satisfying every goal and preservation constraint. T5 tracks endpoints and updates across four to six states. T2 futures and T4 candidate outcomes remain hidden; other tasks expose the relevant states (Tab. 6).

Blender executes prescribed articulation and Bullet dynamics, records states and interactions, and labels predicates and participation. Each T4 candidate resets to the same initial state. Automated acceptance checks execution, identity, physics, and label/input consistency independently of model answers. Inputs contain permitted images, actions, declarations, queries, and schemas; hidden outcomes and mesh-level scoring evidence remain private.

The test set contributes 77,303 answer items, including 27,664 T3/T5 updates; 76.5% of these updates are preservation. Full label counts, execution settings, and the two unresolved annotations appear in the supplement (Tab. 15 and Appendix D).

4. RAVEL: A Spatial Coding Harness

4.1. Model Space in Code

RAVEL models spatial reasoning through executable declarations (Figure 3). From a public initial-construction recipe and declared actions, the harness builds `SceneSpec`, `ActionProgram`, queries, history, and evidence. `SceneSpec` separates object origins and initial geometry from currently valid attributes in a shared reference frame. Queries name predicates and their attribute dependencies: A–B height reads centers; A–B contact reads surfaces.

The harness interprets action programs to reconstruct current attributes from the recipe and action history. Fixed objects retain geometry; a prescribed pure translation updates a valid center as $c_{t+1}(o) = c_t(o) + \Delta_t$. Rotation and free evolution of unheld objects invalidate their centers, leaving initial geometry historical and `current_geometry` absent. Joint operations update de-

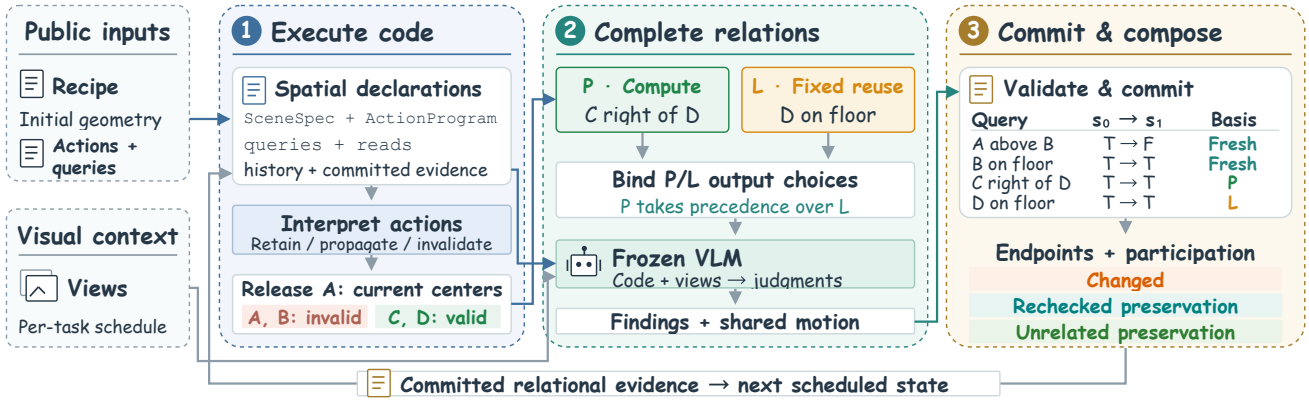


Figure 3. **RAVEL execution loop.** **(1)** Public recipes, actions, and queries become spatial code; interpreting actions reconstructs valid attributes. **(2)** Program findings (P) and eligible fixed evidence (L) bind typed output choices. The frozen VLM reads code and permitted views to complete relations and shared motion. **(3)** The harness validates and commits findings, then composes endpoint values and participation into joint answers. Committed relational evidence carries forward to the next scheduled state. The example follows the Astra collision trace in Tab. 7; T/F denote true/false.

cleared opening coordinates. Python declarations expose this state to the frozen model; the interpreter does not simulate free-body trajectories. Initial centers are authored AABB centers transformed into the question frame, supplied additionally in the main comparison and equally to matched Direct.

4.2. Evaluate Relational Queries

Program findings (P). Direction and proximity execute when both current centers are valid and stored error bounds permit a threshold decision. Declared or prescribed joint coordinates establish opening; authored joint topology establishes attachment. Program findings bind the structured output and take precedence over retained model evidence. After A's release, C–D direction remains computable; A and B lose authoritative centers.

Fixed model evidence (L). A known finding is retained when all named operands are fixed. It retains its original state and evidence basis, as with D–floor support in the collision example. Fixed dependencies establish continued applicability, but an incorrect original finding remains incorrect when reused.

Fresh model judgments. The model resolves remaining relations using permitted images, spatial code, and predicate-specific vocabularies (Tab. 8). Contact, support, and containment require surface or interior evidence beyond centers and bounding boxes. The harness projects findings to true, false, or null. For participation, direct targets are controlled and fixed objects still; the model judges secondary motion of other objects using a 0.025 m center-displacement threshold. Shared participant status then enters every relevant update.

4.3. Compose Joint Decisions

For T3/T5, committed endpoint values and participation determine updates by Equation (1). In the collision trace, A–B height/contact change, B–floor support is rechecked preservation, and fixed C–D direction is unrelated preservation. T1 merges both preservation classes. For T4, let $\mathcal{G}$ contain goal query–value pairs, $\mathcal{K}$ preservation queries, and $\hat{v}_{c,q}$ the prediction under independent candidate c:

$$
\begin{aligned}
G(c) &= \bigwedge_{(q,b)\in\mathcal{G}} [\hat{v}_{c,q} = b], \\
K(c) &= \bigwedge_{q\in\mathcal{K}} [(\hat{v}_{0,q} \neq \bot) \wedge (\hat{v}_{c,q} = \hat{v}_{0,q})], \\
\widehat{\mathcal{C}} &= \{c \in \mathcal{C} : G(c) \wedge K(c)\}.
\end{aligned}
\tag{2}
$$

Every accepted edit achieves all goals and preserves every specified known truth, including false values. Composition enforces these rules over predictions; wrong findings can still yield wrong answers.

4.4. Execute, Complete, and Commit

Each call exposes the spatial code and binds already established P/L findings to singleton output choices; unresolved queries retain their typed vocabularies. This interface carries the representation's execution results into model decoding. The frozen VLM returns relation observations and shared motion judgments; the harness validates, commits, and projects them into joint answers. The model updates relational evidence, not numerical 3D geometry. T1/T3 call once per endpoint; T5 calls once per observed state.

Table 2. **Full-cohort results (4,000 questions; 77,303 answer items).** Metrics are percentages; ΔWhole is RAVEL minus Direct in points. Bold marks the better Whole, Items, T5, and F1 within each backend. Joint accuracy requires both endpoints and the T3/T5 update. RAVEL adds declarations and uses 8,995 calls versus 4,000; matched controls follow in Tab. 3.

		Overall			Whole by task					Changed	Joint endpoints + update		
Backend	Method	Whole	Items	ΔWhole	T1	T2	T3	T4	T5	F1	Changed	Recheck.	Unrel.
Qwen3-VL-4B	Direct	3.60	44.66	–	14.38	2.25	0.00	1.83	0.00	18.85	5.18	16.94	38.50
	RAVEL	**23.40**	**78.92**	+19.80	67.75	10.50	12.60	23.17	**5.63**	**62.40**	50.36	62.68	87.98
Spatial-SSRL-4B	Direct	1.78	37.59	–	8.00	0.13	0.00	1.00	0.00	12.14	2.04	7.96	39.52
	RAVEL	**18.88**	**75.74**	+17.10	62.00	6.75	7.30	17.67	**3.25**	**56.06**	43.28	57.56	87.39
Molmo2-ER	Direct	8.75	47.45	–	39.50	2.75	0.00	2.00	0.00	38.03	0.49	39.84	0.00
	RAVEL	**22.10**	**74.67**	+13.35	65.50	11.13	9.20	22.50	**5.50**	**53.13**	35.33	62.34	78.84
Sol	Direct	62.05	92.76	–	88.13	55.38	55.20	64.00	49.75	88.13	81.63	93.08	94.84
	RAVEL	**79.00**	**96.64**	+16.95	96.00	69.00	77.80	80.83	**72.13**	**95.49**	93.43	96.81	97.64
Astra	Direct	62.70	92.83	–	87.50	57.38	54.30	66.50	50.88	88.25	81.60	93.73	94.30
	RAVEL	**81.20**	**96.96**	+18.50	95.75	73.75	79.10	86.50	**72.75**	**95.71**	94.30	96.78	97.54

Observed-state calls use current and adjacent previous two-view images; earlier actions and findings persist in the code context. T2 advances the context symbolically and makes one initial-view call; T4 makes one initial-view call over four independently initialized branches. Free-body outcomes remain model predictions after numerical premises are invalidated. No hidden endpoints or gold participation enter inference. The same rules apply to all backends; Appendices B.2 and C and Tab. 7 give pseudocode, actual declarations and requests, and the expanded trace.

5. Experiments

5.1. Protocol and Full-Configuration Results

We evaluate on the 4,000-question test set, with 800 questions per task. We evaluate Qwen3-VL-4B-Instruct [1], Spatial-SSRL-4B, Molmo2-ER, and API backends Sol (`gpt-6.1-sol`) and Astra (`gpt-6-astra`), both at medium reasoning effort. Direct receives the question, permitted images, common semantics, and final schema in one call. RAVEL additionally receives public initial geometry and declarations and uses state-wise inference. Both use constrained structured output without answer repair. All 40,000 main answers have valid format; differences concern semantic correctness.

Whole-question accuracy requires every semantic item to be correct; item accuracy micro-averages individual decisions. T4 Whole requires the exact admissible set. For T3/T5, changed F1 scores change labels, while joint accuracy requires both endpoints and the update label. We also decompose T5 into state-vector and complete-trajectory correctness. Paired 95% bootstrap intervals resample source groups for the fixed recorded runs; they do not estimate repeated-decoding variability (Appendix A.3).

The full configuration improves Whole by 13.35–19.80 points across all five backends (Tab. 2). Local models reach 18.88–23.40% Whole and 74.67–78.92% Items; Sol and Astra reach 79.00% and 81.20% Whole. Astra corrects 851 Direct errors and regresses on 111 questions, with a net gain interval of $[16.29, 20.78]$ points. This comparison measures the combined effect of additional declarations, state-wise calls, and executable organization. We next match access to examine the latter more directly.

5.2. Organization under Matched Access

Balanced500 samples metadata across all 100 source groups, with 100 questions per task and 20 per task–physics cell. Each control uses identical raw initial information, legal images, 1,100 state-scheduled calls, and output caps of 1,536 tokens for Qwen and 8,192 for Astra. Matched Direct predicts semantic decisions with a controller that stores and assembles them. Qwen's full-RAVEL reference reuses request-matched frozen responses; Astra's is rerun contemporaneously. Development examined 1,000 main questions and exposed 99 source groups; balanced500 contains 140 examined questions (Appendices A.2 and D.1).

Under this match, Qwen improves from 6.20% to 24.40% Whole, a gain of 18.20 points with interval $[13.98, 22.55]$. Astra improves from 78.00% to 80.20%, a 2.20-point gain with interval $[0.00, 4.57]$, while changed F1 decreases from 95.52% to 94.57%. Equal raw access and calls therefore leave a substantial Qwen benefit and a smaller, mixed Astra outcome (Tab. 3 and Figure 4). Actual tokens and time also differ (Sec. 6.4).

Compulsory binding. We cross program binding P and the fixed-evidence lock L: C00 disables both, C10 enables P, C01 enables L, and C11 is full RAVEL. All cells retain computed hints, typed findings, earlier evidence, and deterministic projection. Qwen gains 15.40 points from

Table 3. **Matched controls on balanced500.** Each row uses 1,100 calls. P/L enforce program findings/fixed evidence; C00–C11 retain hints and history. Cons. invalidates non-fixed centers after every action. Bold: column best. Colored Δ: signed point differences, not significance; brackets give paired 95% intervals. Contrasts are rounded separately.

			Qwen3-VL-4B			Astra		
Configuration	P	L	Whole	Items	F1	Whole	Items	F1
Direct	–	–	6.20	52.64	23.65	78.00	96.14	95.52
C00	0	0	21.60	78.56	62.15	80.00	**96.73**	**95.53**
C10	1	0	23.20	79.21	63.82	78.40	96.53	94.66
C01	0	1	21.40	78.86	62.64	79.40	96.59	95.03
C11: RAVEL	1	1	**24.40**	**80.03**	**65.50**	**80.20**	96.59	94.57
Cons. geometry	1	1	20.40	–	57.45	79.20	–	94.71
ΔWhole (C11−Direct)			+18.20 [13.98, 22.55]			+2.20 [0.00, 4.57]		
ΔF1 (C11−Direct)			+41.86 [35.25, 47.93]			-0.94 [-2.64, 0.55]		

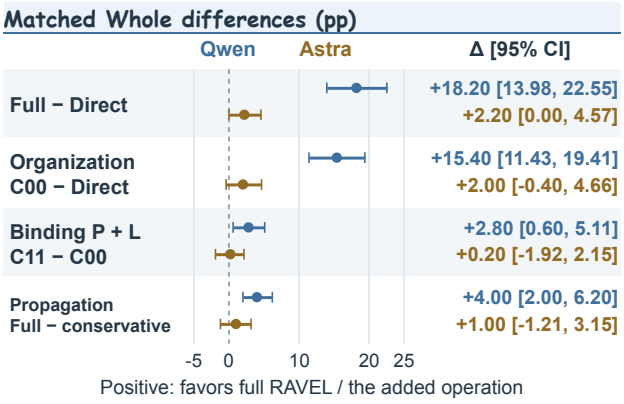


Figure 4. **Organization, binding, and propagation.** Paired Whole differences with source-group bootstrap 95% intervals on balanced500. Qwen is blue; Astra orange. C00 retains computed hints and history. The organization contrast jointly changes several mechanisms; all conditional contrasts are in Tab. 23.

matched Direct to C00, then 2.80 $[0.60, 5.11]$ from C00 to C11. The first contrast changes an organization bundle and does not isolate a single component. With L active, adding P gives 3.00 points; without P, adding L gives -0.20. Astra's C00/C11 scores are 80.00/80.20, with binding-effect intervals spanning zero. When P is disabled, Astra follows all 2,169 hints in both cells; Qwen overrides 26/25. Forced binding thus adds little when a backend already follows the available measurements.

Geometry propagation. Conservative invalidation discards all non-fixed centers after every action, including legal pure translations. Restoring selective propagation raises Qwen Whole from 20.40% to 24.40%, a 4.00-point difference $[2.00, 6.20]$, and changed F1 from 57.45% to 65.50%. Astra gains 1.00 point $[-1.21, 3.15]$, with a small F1 decrease. Retaining action-supported geometry helps Qwen compared with returning those judgments to the model; the marginal effect remains backend-dependent. Complete conditional contrasts appear in Tabs. 23 and 24.

6. Analyzing Joint Relation Judgments

6.1. Changes, Preservation, and Complete Trajectories

The joint scores in Tab. 2 connect complete-answer gains to relation-level judgments. Qwen's rechecked/unrelated joint accuracy rises from 16.94/38.50% to 62.68/87.98% on the full cohort. On matched balanced500, changed/rechecked/unrelated joint accuracy rises from 2.29/31.63/19.20% to 52.00/67.27/88.53%. False change alarms fall from 21.04% to 5.26% for rechecked relations and from 14.58% to 0.24% for unrelated ones, although changed recall remains 53.57%. Thus Qwen improves both detecting change and retaining correct preserved values. Astra's matched joint scores remain nearly unchanged (Tabs. 29 and 30). Aggregate gains also conceal trade-offs: full-cohort Molmo changed recall falls from 39.91% to 38.59% as precision rises from 36.32% to 85.24%.

Complete T5 answers remain harder than individual decisions. On 800 full-cohort trajectories, RAVEL reaches 72.13/72.75% Whole for Sol/Astra and 3.25–5.63% for local models. Astra gets the entire initial vector correct in 728 trajectories, but only 582 have every state and update correct. On 100 matched T5 trajectories, Qwen improves initial-vector accuracy from 3% to 60%; with RAVEL, nine have every state vector correct, twelve every update correct, and eight both. Astra reaches 97.29% Items but 71% Whole, versus 70% matched Direct. These are joint-correctness measurements; identifying retained errors requires provenance, not accuracy alone (Tabs. 16 and 26).

6.2. Validity and Correctness of Evidence

Each full Qwen/Astra run records 55,439 findings, of which 46,439 align with relation-value annotations; 9,000 T4 branch findings lack that alignment. Fresh model findings are correct in 64.51/95.08% of scorable instances for Qwen/Astra, and fixed reuse in 83.44/95.94%. All 14,881 scorable program findings per backend agree with annotations. These source-specific rates cover different queries and states (Figure 5a). Qwen records 917 wrong reuses from 428 initial sources; Astra records 224 from 114. Fresh judgments also yield 9,236/1,281 errors, including 3,118/394 transitions from a previously correct finding (Tab. 35). Continued applicability and initial correctness are distinct properties of evidence.

Retention horizon. On the 100 matched T5 trajectories, we cap original-source age at $h \in \{0, 1, 2, \infty\}$ in both persistent and previous-state channels; copying does not reset

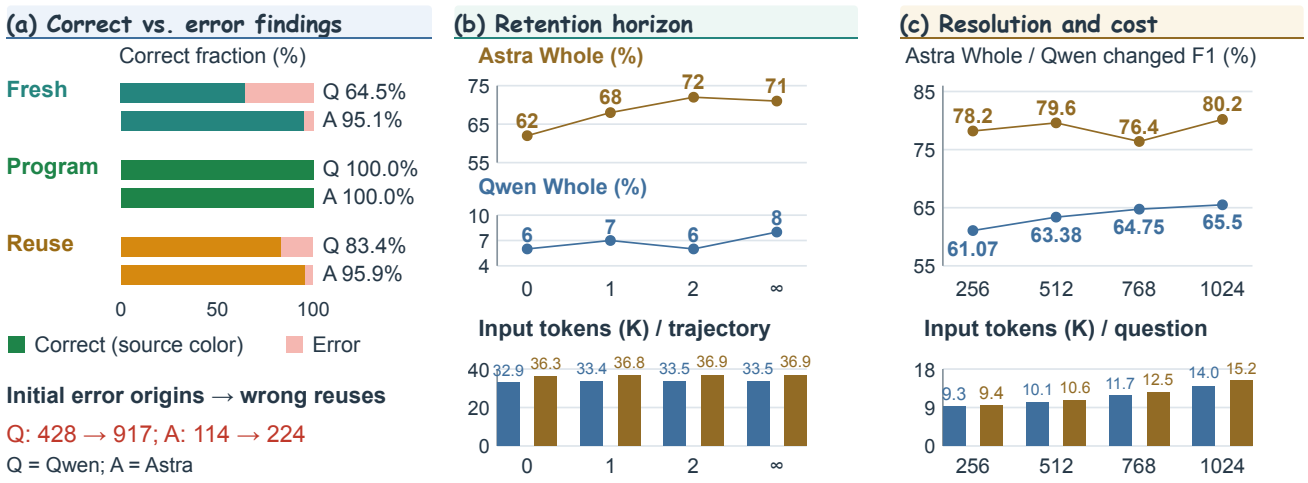


Figure 5. **Evidence quality and sensitivity.** (a) Correct/error fractions among scorable full-run findings; Qwen/Astra counts differ for fresh (26,022/26,039) and reused findings (5,536/5,519), with 14,881 program findings each. (b) T5 Whole versus source-age cap on 100 matched trajectories. (c) RAVEL resolution on balanced500. Blue: Qwen; orange: Astra. Quality curves use local ranges; token bars start at zero. Curves show point estimates; paired-difference intervals appear in Tabs. 25 and 27.

age. Images, actions, initial declarations, and numerical evaluation remain available. Qwen Whole is 6/7/6/8%; Astra is 62/68/72/71% (Figure 5b). Astra's $h = 0$ difference from default is -9.00 points $[-20.18, 2.22]$. Longer retention has no monotonic advantage in these runs; complete intervals and costs appear in Tab. 25.

Recorded traces. In the translation case, Astra Direct keeps A–C proximity true throughout; RAVEL correctly tracks its alternating values with valid numerical geometry (Figure 6a). In the tray-rotation case, Direct is correct but RAVEL repeatedly misses B–floor support after rotation (Figure 6b). These are fresh judgments about a moving operand, so the error is not fixed-evidence reuse. A separate Qwen support trace (R5) retains an error: an incorrect initial finding for fixed D is reused at states 1–3; an Astra trace (R1) provides another instance. The records distinguish successful propagation, repeated fresh misjudgment, and retained initial error.

6.3. Visual and Physical Sensitivity

Unseen physical outcomes remain difficult: local RAVEL backends reach only 6.75–11.13% T2 Whole, versus 69.00–73.75% for API backends. Matched Qwen improves direction-value accuracy from 46.49% to 85.68%, while support changes from 70.44% to 69.69%; its collision Whole is 9%. Astra's matched joint-motion Whole falls from 59% to 57%, with slight decreases in contact and proximity. Full task, physics, predicate, participation-role, and source breakdowns appear in Appendices F and G.

Resolution. Resizing permitted images to 256/512/768/1,024 pixels preserves aspect ratio and all other RAVEL settings. Qwen Whole stays within 23.00–24.60%, but changed F1 falls from 65.50% at 1,024 to 61.07% at 256. Astra Whole is non-monotonic: 76.40% at 768 versus 79.60% at 512. Lower resolution reduces input tokens while affecting metrics differently (Figure 5c,Tab. 27). Direct is not rerun at these resolutions.

Appearance and sources. We evaluate both matched methods under one alternative lighting and one material configuration on balanced500, keeping physical states, cameras, labels, and boxes fixed. Qwen's RAVEL Whole is 24.00/24.20%, with 19.00-point gains in both conditions. Astra gains 1.40 $[-1.40, 4.24]$ and 2.80 $[0.39, 5.38]$ points (Figure 7a). Paired changes in the gain relative to original images have intervals spanning zero. Both source families also improve in the full comparison: Astra gains 17.80 points on HSSD and 22.19 on Infinigen. The source slices overlap development; appearance tests isolate rerendering of existing episodes (Tabs. 12 and 28).

6.4. Accuracy and Inference Cost

The full configuration uses 8,995 calls per backend versus 4,000 for Direct. For Astra, mean input tokens increase from 5,626 to 15,460 and cumulative inference time from 22.18 to 33.92 seconds per question (Figure 7b). Time sums recorded calls within each question; hardware and API conditions differ across backends. Under matched 2.2 calls per question, Qwen uses 11,106/14,018 input tokens and 3.94/13.59 seconds for Direct/RAVEL; Astra uses 11,817/15,250 tokens and 20.77/24.22 seconds. Equal ac-

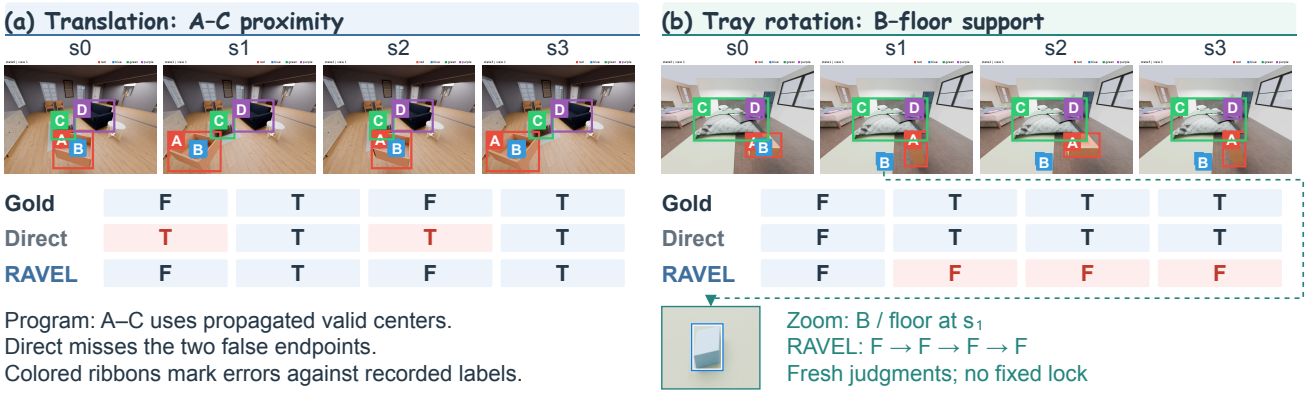


Figure 6. **Recorded correction and regression across four states.** (a) Astra RAVEL correctly alternates A–C proximity using valid geometry. (b) Its fresh visual findings miss B–floor support; Direct is correct. Red ribbons mark errors. The inset magnifies the actual state-1 image; view 1 is shown, while calls use both views. Selection is documented in Appendix E.

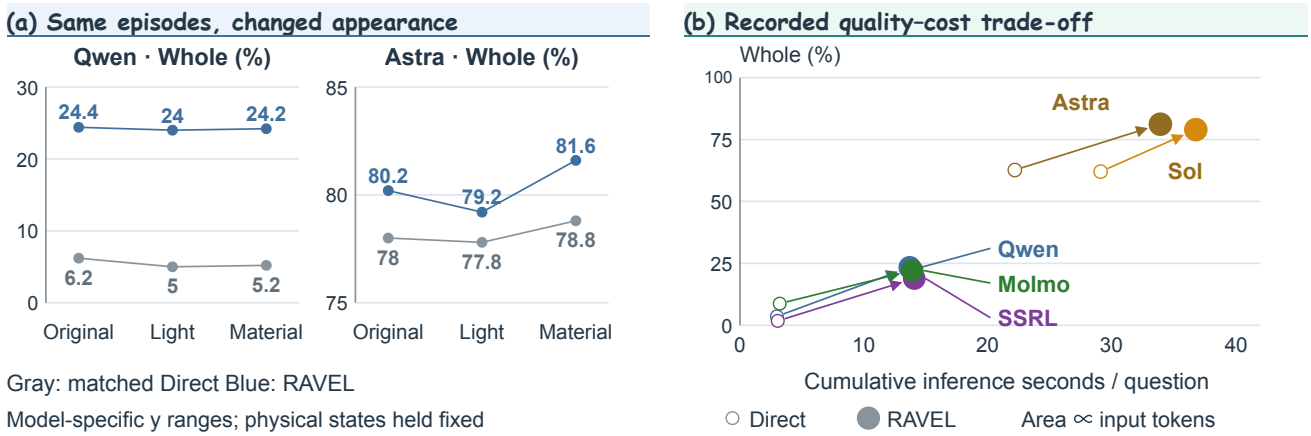


Figure 7. **Appearance robustness and recorded cost.** (a) Both matched methods use the same 500 physical episodes under original, alternative-lighting, and alternative-material renders; y-axis ranges differ by backend. (b) Five-backend full-cohort results: arrows connect Direct to RAVEL; bubble area is proportional to input tokens. Time sums calls per question; differing hardware/API conditions preclude a common hardware efficiency ranking.

cess and schedules therefore leave additional context and generation costs (Tabs. 17 and 34).

7. Limitations and Conclusion

The benchmark uses synthetic indoor scenes, four named objects, and identity boxes. Development exposed 99 of 100 source groups; independent source transfer and human audit remain pending, with two unresolved joint annotations. RAVEL requires precise initial declarations and a finite set of supported operations; fixed locks can retain initial errors. Matched controls cover two backends, and bootstrap intervals describe fixed runs without multiple-comparison adjustment. External benchmark comparisons and repeated full runs are unmeasured. The supplied kit supports offline scoring and seven complete examples; full inference reproduction additionally requires the complete images, assets, weights, and environment.

We introduce a benchmark and inference harness for joint change-and-preservation reasoning under physical actions. RAVEL models space in code: executable declarations link action-dependent attribute validity to computed, fresh, and retained findings, from which the harness composes complete relational answers. The full configuration improves all five backends; matched controls retain a large Qwen gain and a smaller Astra gain. Qwen's improvements span changes and preservation, with its largest increment in executable organization. Unresolved judgments and retained initial mistakes continue to limit complete trajectories.

LLM Usage Statement

OpenAI Codex (GPT-6-based) assisted with drafting and revising text throughout the manuscript, organizing the presentation of supplied experiments, checking related-work references, and writing LaTeX and editable figure-generation code. This assistance included substantive restructuring and compression, table assembly from supplied results, and diagram and plot layout. Scene images in the figures are display-adjusted copies or labeled crops of recorded renders; the figure-generation workflow does not synthesize experimental scenes or model responses. The vision-language models evaluated as research systems and their inference configurations are described in the experimental protocol and appendix.

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A. Experimental Details

Dataset and test-set scale. The dataset contains 12,000 questions, including a 4,000-question test set with 800 questions per task. All main experiments use this test set; balanced500 is a subset with 100 questions per task. Reported answer-item counts, update distributions, source-group statistics, model calls, and accuracy denominators describe the evaluated test set.

A.1. Backends and Baseline Interfaces

All main-table results come from evaluating the frozen question list. Local experiments use the weight identifiers Qwen3-VL-4B-Instruct, Spatial-SSRL-Qwen3VL-4B, and Molmo2-ER. API experiments record requested and returned service identifiers and use medium reasoning effort. These identifiers describe access to the evaluated services; architecture, parameter counts, and training sets are not independently established by the records. Table 4 reports the inference settings.

Direct baseline. The direct request preserves the complete original question, including object identities, reference frame, physical conditions, edits or candidates, query statements, and the final output schema. It includes all task-permitted RGB images in a single call. Both conditions receive the common geometric-support semantics: support is bottom contact or proximity to the named upward surface, and external holding does not decide that relation. Structured decoding constrains schema shape and allowed types. It does not impose gold semantic values on the direct response.

T1 and T3. Direct answering receives both initial views and both post-action views together. RAVEL first commits state-0 findings, then processes state 1 with the current and previous views, the action, and prior evidence. T1 scores one before value, one after value, and one three-way transition; T3 scores each queried before value, after value, and four-way update. The state-wise schedule changes the inference protocol while preserving the union of permitted images.

T2 and T4. Both tasks permit only the two initial views. T2 requires post-action values without displaying the endpoint. In T4, each candidate begins at state 0, and each branch context is produced by cloning the public initial state before applying that candidate's action program. Branch values remain predictions whenever free dynamics invalidate geometry. Candidate selection uses known initial preservation values and the predicted branch values. Private renders of candidate outcomes never enter either condition's prompt.

T5. The direct baseline receives all two-view states in a single request. RAVEL proceeds from state 0 to the last state, supplying current and adjacent previous images at each call. Its committed relation vector is used for the next transition, and fixed-dependency evidence remains in a separate persistent record. The state counts sum to $270 \cdot 4 + 265 \cdot 5 + 265 \cdot 6 = 3,995$ T5 calls. Adding two calls for every T1/T3 question and one for every T2/T4 question yields $8,995$ calls per model.

A.2. Controlled Comparison Protocol

The main direct and RAVEL conditions differ in raw initial geometric information, call schedule, intermediate output organization, and deterministic computation. The follow-up controls use the fixed balanced500 subset and compare both methods with the same raw information and schedule. Every row of Tab. 21 uses those 500 questions; the original 4,000-question full-RAVEL scores are reported separately in Tab. 36. Earlier 1,000-question and selected 200-question development studies have different protocol identities and are excluded from these controls. Appendix G reports the fixed sampling and all follow-up results.

Direct reasoning with the same raw information. This control matches RAVEL's raw initial information: public centers, bounds, construction and physical declarations, and action parameters. The model directly produces Boolean or null relation values, update labels, and T4 candidate decisions, with intermediate predictions available at the corresponding states. The controller stores and assembles those decisions without adopting evaluator-produced relation findings or performing RAVEL's semantic projection. This comparison measures the executable organization bundle under matched raw information access.

Schedule and output matching. The same-information control uses two calls for T1/T3, one for T2/T4, and one per state for T5, with RAVEL's allowed image grouping and per-call output caps. Scheduled calls and maximum output budget match within each backend, giving 1,100 calls per 500-question condition. Qwen uses 1,536 output tokens per call and Astra 8,192, with medium reasoning effort for Astra. Realized input and output tokens are recorded separately because context and generated answers differ in length.

Removing program binding. P off retains valid geometry, action propagation, query declarations, program-measurement hints, and the state-wise schedule. It removes the singleton observation constraint that forces adoption of each program finding, allowing a model choice from the corresponding vocabulary. The intervention measures

Setting	Qwen3-VL-4B	Spatial-SSRL-4B	Molmo2-ER	Both API backends
Temperature	0	0	0	Service default
Random seed	20271004	20271004	20271004	Not specified
Context limit	32,768	32,768	16,384	Service-managed
Maximum pixels	786,432	786,432	786,432	High image detail
Maximum crops	4	4	4	Service-managed
Direct output limit	2,048	2,048	2,048	8,192, including reasoning
RAVEL output limit/call	1,536	1,536	1,536	8,192, including reasoning
Reasoning effort	N/A	N/A	N/A	Medium
Concurrent API workers	N/A	N/A	N/A	2
External tools / repair	None / none	None / none	None / none	None / none

Table 4. **Main 4,000-question inference settings.** Numeric output limits are token caps per call. Service-managed indicates a setting not specified by the client, rather than a measured service property. API output accounting includes reasoning tokens. Local memory fraction is 0.8 with one sequence per worker. Follow-up matched controls use the RAVEL per-call caps for both methods.

forced adoption of available computation, not removal of the computation itself.

Removing persistent finding locks. L off preserves earlier model findings and their provenance while removing the singleton constraint for eligible fixed-dependency observations. Crossing P and L gives C00, C10, C01, and C11; all four retain derived state, typed findings, and deterministic answer composition. Program binding takes precedence where both constraints apply. With P off, Qwen overrides 26 of 2,169 hints in C00 and 25 in C01; Astra follows all 2,169 in each condition.

Representation scope. RAVEL renders its public context as executable Python declarations. The matched and binding comparisons retain this serialization; an information-equivalent JSON or prose control has not been run. An advantage attributable specifically to code syntax remains untested.

A.3. Metrics and Aggregation

Let m_i be the number of scored semantic items in question i, and let e_{ij} indicate exact typed agreement between prediction and annotation for item j. Booleans, null, and update strings are distinct types; an integer is not a Boolean. Whole correctness is $w_i = \prod_{j=1}^{m_i} e_{ij}$. The two main metrics are

$$\text{Whole} = \frac{1}{N}\sum_i w_i, \qquad \text{Items} = \frac{\sum_i \sum_j e_{ij}}{\sum_i m_i}. \tag{3}$$

The main-comparison denominator is $N = 4,000$ questions or $\sum_i m_i = 77,303$ answer items per condition. Balanced500 uses 500 questions and 9,169 items; its T5 subset contains 100 trajectories and 5,824 items. The complete final output is scored, including incorrect but valid null predictions.

The item metric weights questions by their number of answers. A question-macro alternative first computes $(1/m_i)\sum_j e_{ij}$ and averages over questions. A task-macro whole score averages the five task-specific whole accuracies rather than pooling all questions. The frozen tables retain these alternatives, while the main table uses pooled whole and item accuracy. T1 contributes 2,400 items, T2 5,283, T3 19,875, T4 2,400, and T5 47,345.

Changed precision, recall, and F1 treat changed as positive and the other three update labels as non-positive. Thus $P = TP/(TP + FP)$, $R = TP/(TP + FN)$, and $F_1 = 2PR/(P + R)$, with zero used when a denominator is empty. For a gold update class c, joint correctness averages

$$J_{t,q} = \mathbf{1}[\hat{v}_{t,q} = v_{t,q}]\mathbf{1}[\hat{v}_{t+1,q} = v_{t+1,q}] \cdot \mathbf{1}[\hat{u}_{t,q} = u_{t,q}] \tag{4}$$

over instances whose gold label is c. A predicted label can be correct while $J_{t,q} = 0$ because an endpoint is wrong.

For T4, let $\mathcal{C}_i$ be the four candidates, G_i the annotated valid set, and $\hat{G}_i$ the predicted set. RAVEL assembles $\hat{G}_i$ through the constraint conjunction in Eq. (2) of the main paper; direct answering predicts the set in its final response. Each candidate contributes the membership item $\mathbf{1}[c \in \hat{G}_i] = \mathbf{1}[c \in G_i]$. Whole correctness requires $\hat{G}_i = G_i$, independent of set ordering. Duplicate or non-candidate identifiers violate the schema, while an empty valid set is legitimate when specified by the annotation.

The independent offline scorer confirms 4,000 final answers in every one of the ten model–condition combinations. All final answers have valid format, with zero final missing outputs, runtime errors, or terminal model-output failures. Some API requests required transport retries af-

ter gateway timeouts; retries continued the frozen question list without repairing semantic answers. A valid format establishes that an output can be scored, rather than that its relation values are correct.

A.4. Executed Scene Construction

Each construction job fixes the task, category, source group, seed, participants, and action parameters. The Blender pipeline establishes metric scale and the ground frame, restores furniture, and constructs collision, support, interior, or articulated structures. It declares fixed and movable objects and their mass, friction, restitution, damping, gravity, and control timing. Articulated parts follow prescribed transforms; Bullet runs free-body motion at 60 frames per second, with 12 substeps per frame and 30 solver iterations. Saved transforms and interaction evidence determine relation endpoints and participation. Each T4 candidate resets the same initial state before execution. Two views per state are rendered, with identity boxes obtained from instance evidence. Acceptance checks cover action execution, participation, box validity, physical and geometric conditions, and label/input consistency. Failed jobs are adjusted or resampled and rerun independently of model performance. The reference frame is fixed across states: x follows initial view 1's right direction projected onto the ground, y points horizontally away from that view, and z is vertical.

B. Predicate and State Semantics

Public query records supply the predicate, operands, reference frame, thresholds, and margins. Private state evidence supplies the evaluated geometry used for annotation. The seven query counts in Tab. 5 count each query once per question, before expansion into repeated state values and update labels. No visibility or third-party occlusion predicate is part of the current seven-predicate evaluation.

Predicate	Queries	Operational definition
`direction`	5,980	Signed center difference on a declared axis; a 0.03 m boundary band around zero is unknown.
`near_center`	2,921	3D center distance below the query threshold; the ± 0.03 m band is unknown.
`on_surface`	9,790	Bottom proximity to upward triangles of the named surface; geometric support, including when an external hold is active.
`contact`	1,572	Evaluated triangle-mesh surface distance with threshold 0.02 m and boundary margin 0.003 m.
`inside`	367	Real object vertices within the registered tray interior, with the stored geometric tolerance.
`is_open`	680	Absolute joint coordinate divided by full travel; above 0.55 is open, below 0.35 is closed, and the intervening band is unknown.
`joint_connected`	669	Part remains attached through the authored native joint, including while open.

Table 5. **Current query families.** Counts are queries, not scored state values. The individual query record is authoritative for its parameters. Surface and interior evidence is used for annotation; RAVEL delegates these predicates to the model when valid public program data cannot resolve them.

Support annotation samples evaluated mesh vertices within 0.008 m of the object's bottom, retaining at most 48. It casts downward rays from 0.03 m above each sample over 0.09 m, accepts surfaces with upward normal component above 0.65, and labels support true when the smallest accepted absolute vertical gap is below 0.018 m. This finite geometric test evaluates bottom proximity rather than force equilibrium. Tray containment transforms real vertices into the tray's registered frame and tests minimum interior clearance against a tolerance of -0.012 m. Both annotation procedures use mesh-level evidence beyond bounding-box overlap.

The public context stores scene declarations in `SceneSpec` and actions in `ActionProgram`. `origin.initial_geometry` preserves state-0 declarations, while `current_geometry` records the maintained numerical geometry with its state and source. The initial center is the authored AABB center transformed into the question frame, rather than an image-box center or a center of mass. Declarations also contain construction parameters or imported categories, body settings, and the floor reference elevation; the latter does not specify the floor mesh. Action declarations provide targets, translations or joint coordinates, timing, hold/release operations, and free-evolution conditions. The creation recipe records the initial construction trajectory; the model-facing spatial code is compiled from that recipe, the declared actions, queries, and committed evidence. For each request, `world_for` rebuilds numerical state by replaying the declared actions from the initial recipe, while committed evidence passes between requests. Thus persistence refers to the logical spatial representation, rather than a resident 3D object modified by the model. Pure translations also propagate applicable vertical bounds; a release waypoint does not establish the final free-body pose, and holding an object cannot restore an already invalidated center. The finite predicate-to-attribute `reads` mapping supplies query dependencies; it is not inferred by general program analysis. The public state engine initializes centers from authored initial AABB centers, with a stored numerical error bound of 10^{-6} m. Each propagated pure translation increases that bound by 10^{-6} m. Directional evaluation checks the distance to the margin against the sum of center error bounds; proximity checks the threshold-band edges with the corresponding $\sqrt{3}$-scaled bound. A query too close to those numerical decision boundaries is delegated rather than force-bound. Fixed objects retain valid centers; rotated targets and freely evolved, unheld participants lose authoritative centers. Program measurements are computed in the current state before the request is rendered.

B.1. Complete Task Interfaces and Recorded Relational Trace

Table 6 specifies each task's question count, permitted images, and scored answer. Table 7 follows the collision example from its recorded relation findings to the final update decisions.

Task	Questions	Permitted images	Scored answer
T1: single update	800	Two views before and after	One relation before/after; three-way update
T2: outcome prediction	800	Two initial views	Multiple post-action relation values
T3: selective update	800	Two views before and after	Multiple endpoint values; four-way updates
T4: constrained edits	800	Two initial views	Set of valid candidates among four independent edits
T5: consecutive updates	800	Two views at each of 4–6 states	Values at every state; all adjacent four-way updates

Table 6. **Task interfaces.** Each task is crossed with all five physical categories, giving 25 cells. A source group can contribute multiple questions. Candidate futures in T4 and the T2 future state are excluded from model input. T5 supplies observations along a recorded sequence rather than an agent-controlled interaction loop.

Relation query	State 0: basis / finding	State 1: basis / finding	Relational conclusion
R1: D on floor	Current images / bottom on named surface	Persistent fixed observation / same finding	True $\rightarrow$ true; unrelated, preserved
R2: A higher than B	Valid numeric geometry / positive z difference	Current images / negative z difference	True $\rightarrow$ false; changed
R4: C right of D	Valid numeric geometry / positive x difference	Valid numeric geometry / positive x difference	True $\rightarrow$ true; unrelated, preserved
R5: A contacts B	Current images / separated	Current images / touching	False $\rightarrow$ true; changed
R7: B on floor	Current images / bottom on named surface	Current images / bottom on named surface	True $\rightarrow$ true; B moves secondarily, so rechecked, preserved

Table 7. **An executed relational inference trace.** Five of seven queries from the collision example in the main-paper teaser, using the recorded `gpt-6-astra` RAVEL responses. After release, A and B lose authoritative current centers, while fixed C–D direction remains computable. The model judges B to have moved secondarily and to remain on the floor; the controller combines these findings into a rechecked-preserved update. Fixed D–floor support retains its original attributed observation. Both direct and RAVEL answers are correct on this example.

B.2. Controller Pseudocode

Algorithm 1 summarizes the harness controller. *Build and bind* prepares the model-facing context and schema using current numerical or joint measurements, then eligible fixed-dependency model findings, and leaves other observations free within their typed vocabulary. *Call* submits the permitted images and context to the backend. *Commit* validates state/schema agreement and records the findings and provenance. The controller projects these findings into the task answer and retains eligible evidence for later scheduled calls. The free dynamics declaration informs the model and invalidation rules; it does not invoke a simulator during inference.

Algorithm 1 RAVEL harness inference

Require: Public episode p, model M
Initialize public state W and empty evidence E
if p is T4 **then**
    Build initial query context from W
    for each independent candidate c **do**
        Clone W into W_c
        Apply c's update/invalidation program to W_c
        Build and bind branch findings for W_c
    end for
    Call M once with initial views and branch contexts
    Commit initial and branch findings
    return candidates satisfying all goals and keep tests
else if p is T2 **then**
    Apply the declared action program to W
    Build and bind predicted-state findings
    Call M once with initial views
    return relation values projected from findings
else
    for each observed state t in order **do**
        if $t > 0$ **then**
            Apply action update/invalidation to W
        end if
        Build and bind findings using W and E
        Call M with current/previous allowed views
        Commit typed observations and motion
        Project values and adjacent update labels
        Retain eligible fixed-dependency evidence in E
    end for
    return the task's assembled state/update answer
end if

C. Concrete Input and Output Interfaces

`prompts/direct.txt` contains the collision example's recorded direct request: shared support semantics, the original question, permitted image references, and the final answer schema. The image bytes are stored separately and displayed in their recorded order.

A recorded spatial representation. Listing 1 preserves selected fields and values from the state-1 Python context for the collision example in the additional Qwen3-VL-4B records. Ellipses omit other declarations and fields; this excerpt is illustrative, not a standalone program. The initial center remains in `origin`, while `current_geometry=None` with-

draws it as a current numerical premise after release. The query explicitly records which attributes its evaluation reads.

Listing 1. Selected spatial declarations after release.

```
# SceneSpec initialization omitted.
scene.object('A',
  origin={'initial_geometry': {
    'center_m': [0.98, -0.0, 0.804957],
    'valid_at': 0}},
  current_geometry=None)
action = ActionProgram.from_spec({
  'before_state': 0, 'after_state': 1,
  'kind': 'release', 'targets': ['A'],
  'steps': [...]})
queries = {'R2': {
  'predicate': {'kind': 'direction',
    'a': 'A', 'b': 'B', 'axis': 2,
    'sign': 1, 'margin': 0.03},
  'reads': ['A.center_question',
            'B.center_question'],
  'decision': 'model_required'}}
```

This declaration exposes an unresolved dependency to the model; it does not assign the relation's truth or reconstruct the missing center. In the same request, R4 uses valid C–D centers and R1 reuses a fixed D–floor observation. These Qwen records serve as interface evidence and are distinct from the successful Astra trace in Tab. 7. Qwen reports B as still and R2 as true at both endpoints, so execution and composition preserve those semantic errors. The five additional recorded cases (12 requests) were replayed offline to verify declarations, request constraints, evidence commits, and answer assembly; replay does not constitute a new accuracy experiment.

The corresponding RAVEL context and full schema are provided in `prompts/`. Table 8 lists the actual observation vocabularies. Each query also has a schema-bound basis, such as current images, action prediction, valid numeric geometry, declared joint data, or persistent fixed observation. The following excerpt is the actual state-1 schema for R4, the direction between fixed C and D in the collision example:

```
{
  "type": "object",
  "properties": {
    "basis": {"enum": ["valid_numeric_geometry"]},
    "observation": {"enum": ["a_positive"]}
  },
  "required": ["basis", "observation"],
  "additionalProperties": false
}
```

Query family	Permitted observations before any singleton binding
Direction	`a_positive`, `a_negative`, `boundary`, `unresolved`
Center proximity	`near`, `far`, `boundary`, `unresolved`
Surface support	`bottom_on_named_surface`, `bottom_on_other_surface`, `clear_gap_to_named_surface`, `boundary`, `unresolved`
Containment	`inside`, `outside`, `boundary`, `unresolved`
Contact	`touching`, `separated`, `boundary`, `unresolved`
Opening	`open`, `closed`, `boundary`, `unresolved`
Attachment	`connected`, `disconnected`, `unresolved`

Table 8. **Typed model findings.** Boundary and unresolved map to null. Other observations map to the predicate's truth value; direction additionally uses the query's declared sign. The three positive/negative support observations map to supported or not-supported before truth projection.

The shared final fields are `before`, `after`, and `transition` for T1; `after` for T2; `before`, `after`, and `update` for T3; `valid_candidates` for T4; and `states` and `transitions` for T5. RAVEL elicits intermediate findings and motion where applicable, then assembles these fields deterministically. The recorded state-1 observations below include findings bound before the call and findings left for the model to infer. Schema validity establishes the structure of these observations; their semantic correctness is assessed against the annotation.


```
{
  "findings": {
    "R1": {"basis": "persistent_fixed_observation",
           "observation": "bottom_on_named_surface"},
    "R2": {"basis": "current_images", "observation": "a_negative"},
    "R3": {"basis": "current_images", "observation": "bottom_on_named_surface"},
    "R4": {"basis": "valid_numeric_geometry", "observation": "a_positive"},
    "R5": {"basis": "current_images", "observation": "touching"},
    "R6": {"basis": "current_images", "observation": "near"},
    "R7": {"basis": "current_images", "observation": "bottom_on_named_surface"}
  },
  "motion": {
    "A": "controlled",
    "B": "secondary_moved",
    "C": "still",
    "D": "still"
  },
  "state": 1
}
```


D. Audit Status and Development Exposure

The 4,000-question production list has completed automated generation acceptance. Recorded formal assistant visual reviews and formal human reviews both number zero. Earlier inspection of development examples is excluded from these independent audit counts. Blank outcomes in Tab. 9 mark reviews that have not been performed.

Check	Sample	Result
Reference runtime	4,000	3,998 pass
Formal assistant visual review	0	—
Formal human visual review	0	—
Independent audit (planned)	250–500	—

Table 9. **Current audit record.** Planned review size is a protocol range, not a completed sample. The two unresolved reference cases have not been adjudicated as annotation errors.

The planned independent audit will sample 250–500 questions stratified across task–physics cells, source families, relation types, and update labels. Reviewers will inspect permitted images, action and query semantics, and the relevant hidden execution evidence to verify annotations. They will record object correspondence, visible geometry, threshold interpretation, update scope, and unresolved cases. This protocol assesses annotation quality and input permissions; a human task-solving baseline would require a separate study.

The reference runtime leaves two joint-motion cases unresolved because of relation or update disagreements. The cause remains unadjudicated, and both questions retain their frozen labels and scores in the reported cohort. Resolving the disagreements requires tracing the geometry, joint state, and label computation before deciding whether to issue a versioned annotation correction.

The research kit packages the complete main-comparison question records, frozen annotations, 40,000 final answers, intermediate records, aggregate tables, generation and inference code snapshots, and seven complete image/request examples. Its independent scorer recomputes main-comparison results without importing the original scorer. The follow-up tables and figures use the author-supplied aggregate results; their raw response and bootstrap archives are separate from this kit. Re-

running inference additionally requires the remaining images, scene assets, local weights or API access, and a resolved execution environment.

Model	Condition	Direct	Indirect	Unrelated
Qwen3-VL-4B	Direct	12.31	13.86	38.28
	RAVEL	61.93	23.32	88.08
Spatial-SSRL-4B	Direct	5.63	6.41	39.25
	RAVEL	56.08	17.88	87.50
Molmo2-ER	Direct	26.84	9.51	0.00
	RAVEL	54.24	33.04	79.02
Sol	Direct	88.48	86.90	94.09
	RAVEL	94.70	97.83	97.66
Astra	Direct	89.04	85.98	93.54
	RAVEL	94.96	98.53	97.56

Table 10. **T3/T5 joint correctness by recorded participation role (%).** Role labels are private offline analysis metadata. Joint correctness requires both endpoints and the update label; roles do not identify counterfactual causal effects.

D.1. Development Exposure

Table 11 separates the 1,000 questions examined during development from the 3,000 not individually examined. The indicator concerns prior question-level analysis, not a disjoint room split. Ninety-nine of the 100 source groups were already present in development. The source-family table and this slice therefore describe behavior under shared source exposure. A new-source evaluation must define and acquire its own disjoint source collection before providing a transfer estimate.

	Examined (1,000)		Not examined (3,000)	
Model	Direct	RAVEL	Direct	RAVEL
Qwen3-VL-4B	3.30	22.50	3.70	23.70
Spatial-SSRL-4B	1.20	17.80	1.97	19.23
Molmo2-ER	8.50	21.10	8.83	22.43
Sol	61.10	77.50	62.37	79.50
Astra	61.50	81.20	63.10	81.20

Table 11. **Development-exposure slices: whole accuracy (%).** Examined denotes prior individual-question analysis. The unexamined questions share source groups with development; 99 of 100 groups had been exposed.

	HSSD		Infinigen	
Model	Direct	RAVEL	Direct	RAVEL
Qwen3-VL-4B	3.72	23.75	2.97	21.56
Spatial-SSRL-4B	1.96	19.76	0.78	14.22
Molmo2-ER	9.11	23.33	6.88	15.63
Sol	61.01	77.44	67.50	87.19
Astra	62.08	79.88	65.94	88.13

Table 12. **Source-family whole accuracy (%).** HSSD contributes 3,360 questions from 80 groups; Infinigen contributes 640 from 20 groups. These are current-cohort slices with development source overlap.

E. Case Provenance

Five packaged cases are chosen as the lexicographically first episodes in T1/joint, T2/gravity, T3/collision, T4/surface, and T5/carry, without an accuracy filter. Two additional T5 cases are selected within Astra outcomes: the translation example (Figure 12a) is the first RAVEL-correct/direct-incorrect case, and the tray-rotation example (Figure 12b) is the first direct-correct/RAVEL-incorrect case. Each includes both views at all permitted states, the frozen answer, all ten final predictions, and recorded Qwen/Astra requests and outputs. The draw.io figures embed display copies with uniform gamma 1.18 and contrast 1.10. Existing identity bounds, recovered from the annotation ink, are redrawn as thicker editable boxes. The labeled trajectory inset crops the recorded state-1 image; no instance mask or scene content is synthesized. The original experimental images remain unchanged. The task overview shows one initial view from each physical category, and the trajectory figure shows view 1 at all four states of both selected cases. All permitted views remain in the underlying records. Case-level differences explain actual executions and do not identify the causal effect of an individual component.

F. Paired Uncertainty and Detailed Results

For the two API backends, the paired bootstrap samples 100 source groups with replacement for each of 10,000 replicates using seed 20261014. Both conditions retain the same sampled groups and multiplicities. In each replicate, the whole-accuracy difference is computed from the pooled question counts in those sampled clusters, and the 2.5th/97.5th percentiles form the interval. Table 13 gives the underlying paired outcomes. Each condition has one frozen full run, so these intervals do not measure repeated-run decoding variability. The 100-question API pilot responses are included in the 4,000-question cohort, not additional independent observations.

Model	Both correct	RAVEL only	Direct only	Both wrong
Qwen3-VL-4B	103	833	41	3,023
Spatial-SSRL-4B	51	704	20	3,225
Molmo2-ER	263	621	87	3,029
Sol	2,351	809	131	709
Astra	2,397	851	111	641

Table 13. **Paired whole-question outcomes.** Each row partitions the same 4,000 questions. API confidence intervals resample source groups.

Model	Condition	T1	T2	T3	T4	T5
Qwen3-VL-4B	Direct	14.38	2.25	0.00	1.83	0.00
	RAVEL	67.75	10.50	12.60	23.17	5.63
Spatial-SSRL-4B	Direct	8.00	0.13	0.00	1.00	0.00
	RAVEL	62.00	6.75	7.30	17.67	3.25
Molmo2-ER	Direct	39.50	2.75	0.00	2.00	0.00
	RAVEL	65.50	11.13	9.20	22.50	5.50
Sol	Direct	88.13	55.38	55.20	64.00	49.75
	RAVEL	96.00	69.00	77.80	80.83	72.13
Astra	Direct	87.50	57.38	54.30	66.50	50.88
	RAVEL	95.75	73.75	79.10	86.50	72.75

Table 14. **Whole-question accuracy by task (%).** Each task has 800 test questions. T2 and T4 provide initial views only. T5 requires every state value and transition label to be correct.

		Changed detection (%)			Joint correctness (%)		
Model	Condition	P	R	F1	Changed	Rechecked	Unrelated
Qwen3-VL-4B	Direct	50.94	11.56	18.85	5.18	16.94	38.50
	RAVEL	79.47	51.37	62.40	50.36	62.68	87.98
Spatial-SSRL-4B	Direct	47.17	6.97	12.14	2.04	7.96	39.52
	RAVEL	76.51	44.24	56.06	43.28	57.56	87.39
Molmo2-ER	Direct	36.32	39.91	38.03	0.49	39.84	0.00
	RAVEL	85.24	38.59	53.13	35.33	62.34	78.84
Sol	Direct	95.67	81.69	88.13	81.63	93.08	94.84
	RAVEL	97.58	93.49	95.49	93.43	96.81	97.64
Astra	Direct	96.04	81.63	88.25	81.60	93.73	94.30
	RAVEL	97.11	94.36	95.71	94.30	96.78	97.54

Table 15. **Change detection and preservation.** T3/T5 contain 6,331 changed, 10,562 rechecked-preserved, and 10,590 unrelated-preserved instances. Joint correctness requires both endpoint values and the update label to be correct. The 181 gold-unknown instances are excluded from these three category columns. Changed P/R/F1 treat all four labels as the prediction space.

Model	Condition	Initial correct	4 states	5 states	6 states	All
Qwen3-VL-4B	Direct	9/800	0.00	0.00	0.00	0.00
	RAVEL	468/800	5.56	4.91	6.42	5.63
Spatial-SSRL-4B	Direct	4/800	0.00	0.00	0.00	0.00
	RAVEL	357/800	2.96	2.64	4.15	3.25
Molmo2-ER	Direct	24/800	0.00	0.00	0.00	0.00
	RAVEL	247/800	4.44	6.04	6.04	5.50
Sol	Direct	535/800	46.30	48.30	54.72	49.75
	RAVEL	726/800	71.48	74.34	70.57	72.13
Astra	Direct	524/800	47.04	52.08	53.58	50.88
	RAVEL	728/800	71.48	75.47	71.32	72.75

Table 16. **T5 whole-trajectory accuracy (%).** Four-, five-, and six-state groups contain 270, 265, and 265 trajectories. Initial correct counts require the entire initial relation vector to match the annotation. Length slices contain different questions and are not a controlled horizon intervention.

Model	Condition	Calls	Input tok./q.	Output tok./q.	Seconds/q.
Qwen3-VL-4B	Direct	4,000	5049.68	84.08	2.98
	RAVEL	8,995	14197.88	440.99	13.70
Spatial-SSRL-4B	Direct	4,000	5049.68	84.17	3.05
	RAVEL	8,995	14197.85	440.91	14.06
Molmo2-ER	Direct	4,000	4983.48	51.44	3.21
	RAVEL	8,995	14097.50	247.04	13.90
Sol	Direct	4,000	5626.11	424.54	29.10
	RAVEL	8,995	15459.94	519.75	36.78
Astra	Direct	4,000	5626.11	225.41	22.18
	RAVEL	8,995	15459.87	379.50	33.92

Table 17. **Recorded inference cost.** Calls are totals over 4,000 questions; tokens and seconds are per-question means. Output tokens follow each backend's accounting; API totals include reasoning tokens. Seconds sum the calls within a question and do not measure concurrent wall-clock completion. Token totals from different tokenizers are not a common computational unit.

Table 19 reports RAVEL whole accuracy in every task–physics cell. Table 18 pools relation-value decisions by predicate, counting repeated states separately and retaining null values. Table 10 groups T3/T5 updates by the private direct, indirect, or unrelated participation role and applies the same joint endpoint-and-label criterion used in the main analysis. These tables decompose the frozen comparison at different units; query counts, relation-value counts, transition counts, and whole-question counts are not interchangeable denominators.

Model	Condition	Direction	Near	On surface	Contact	Inside	Open	Connected
Qwen3-VL-4B	Direct	34.20	36.08	69.48	35.20	66.59	19.87	73.75
	RAVEL	86.06	85.66	69.38	65.10	94.41	100.00	100.00
Spatial-SSRL-4B	Direct	25.35	20.54	63.81	24.52	52.68	8.83	61.61
	RAVEL	86.57	84.99	65.38	45.30	88.94	100.00	100.00
Molmo2-ER	Direct	49.62	51.15	53.84	49.07	70.99	50.17	80.45
	RAVEL	85.87	86.14	57.49	54.94	72.89	100.00	100.00
Sol	Direct	92.66	85.89	92.72	95.74	100.00	99.93	100.00
	RAVEL	98.38	95.94	94.99	95.28	100.00	100.00	100.00
Astra	Direct	92.66	85.88	92.71	96.96	100.00	99.93	100.00
	RAVEL	98.64	96.30	95.26	96.29	100.00	100.00	100.00

Table 18. **Relation-value accuracy (%).** Aggregates relation values in T1/T2/T3/T5, including gold null values. Repeated states count separately. These denominators differ from the once-per-question query counts in Tab. 5.

Table 19. **Physical-category accuracy and task-specific slices (%).** Each model block begins with the Direct and RAVEL aggregate scores, followed by RAVEL scores in all 25 task–physics cells. The five tasks cover all five physical categories.

Model	Condition / task	Joint	Gravity	Surface	Collision	Carry
Qwen3-VL-4B	Direct / all	4.38	4.13	1.00	2.00	6.50
	RAVEL / all	35.63	17.50	17.88	12.00	34.00
	RAVEL / T1	68.75	70.63	59.38	59.38	80.63
	RAVEL / T2	23.75	0.63	0.00	0.00	28.13
	RAVEL / T3	19.50	4.50	8.50	0.00	30.50
	RAVEL / T4	60.00	9.17	25.00	0.83	20.83
	RAVEL / T5	16.25	3.75	0.63	0.00	7.50
Spatial-SSRL-4B	Direct / all	2.38	1.75	0.75	0.50	3.50
	RAVEL / all	33.38	14.50	11.88	10.38	24.25
	RAVEL / T1	70.00	60.63	45.63	51.88	81.88
	RAVEL / T2	22.50	0.00	0.00	0.00	11.25
	RAVEL / T3	18.00	4.50	1.00	0.00	13.00
	RAVEL / T4	54.17	6.67	16.67	0.00	10.83
	RAVEL / T5	11.25	1.25	0.00	0.00	3.75
Molmo2-ER	Direct / all	12.13	8.63	6.75	5.00	11.25
	RAVEL / all	40.00	12.13	13.38	12.13	32.88
	RAVEL / T1	78.75	41.88	65.00	53.13	88.75
	RAVEL / T2	27.50	1.25	1.88	0.63	24.38
	RAVEL / T3	24.50	0.00	0.00	0.00	21.50
	RAVEL / T4	60.83	23.33	0.00	9.17	19.17
	RAVEL / T5	17.50	0.00	0.00	0.00	10.00

Continued on next page

Table 19 (continued)

Model	Condition / task	Joint	Gravity	Surface	Collision	Carry
Sol	Direct / all	47.88	67.75	77.50	52.88	64.25
	RAVEL / all	59.63	89.63	85.25	65.63	94.88
	RAVEL / T1	88.13	99.38	95.00	98.13	99.38
	RAVEL / T2	51.88	92.50	78.13	26.25	96.25
	RAVEL / T3	42.50	90.00	79.00	85.00	92.50
	RAVEL / T4	80.83	88.33	92.50	48.33	94.17
	RAVEL / T5	44.38	77.50	85.00	61.25	92.50
Astra	Direct / all	47.75	67.88	77.13	56.13	64.63
	RAVEL / all	59.50	89.63	86.63	74.75	95.50
	RAVEL / T1	85.00	99.38	97.50	97.50	99.38
	RAVEL / T2	54.38	92.50	81.25	44.38	96.25
	RAVEL / T3	42.00	89.00	81.50	90.50	92.50
	RAVEL / T4	80.00	87.50	92.50	75.00	97.50
	RAVEL / T5	45.63	79.38	83.13	62.50	93.13

G. Follow-up Protocols and Complete Results

G.1. Cohort, Reference Runs, and Uncertainty

Balanced500 is sampled from question metadata before reading model outcomes or annotations, using seed 20261020. It contains 100 questions per task and 20 per task–physics combination, covers all 100 original source groups, and comprises 427 HSSD and 73 Infinigen questions. Both evaluation sets assign equal weight to T1–T5, with 100 and 800 questions per task, respectively. Of its 500 questions, 140 were previously examined individually, and 99 source groups were exposed during development. The subset supports paired mechanism analysis within this cohort.

The 100 T5 questions are complete trajectories from 48 source groups, with 34 four-state, 32 five-state, and 34 six-state trajectories, totaling 500 states. The age interventions retain every state and action. They use 500 calls per condition, while every full balanced500 condition uses 1,100. All conditions preserve identity boxes, initial declarations, actions, and task-specific image permissions. No inference condition calls a physical simulator or uses a model-directed external-tool loop.

Qwen retains the frozen temperature of zero and seed 20271004. Its C11 reference reuses the matching 500 questions after checking 1,100 request records and 500 question submissions. Astra uses the requested service identifier `gpt-6-astra`, medium reasoning effort, and concurrency two, with a contemporaneous C11 rerun for all follow-up contrasts. The older Astra C11 subset scores 78.80% Whole and the contemporaneous run 80.20%. Their difference is 1.40 points with interval $[-0.39, 3.21]$, comprising 14 corrections and seven regressions. The original 4,000-question Astra result remains 81.20%.

Paired intervals resample actual source groups 10,000 times with seed 20261014 and use percentile 95% bounds. The same group multiplicities are used across compared conditions and all four cells of the binding design. Whole, micro-averaged Items, F1, and their contrasts are recomputed within each replicate. Balanced500 has 100 groups and the T5-only age analysis has 48. The intervals describe source-group variation for one recorded run per condition and use no multiple-comparison adjustment.

Table 20 accounts for all new inference: 13,100 question–condition answers and 30,500 logical calls across 31 new conditions. All final answers have valid format, with no missing final output or terminal inference failure. These are repeated evaluations of a fixed question set, not 13,100 independent questions. The four older reference conditions read during analysis bring the aggregate inventory to 35 conditions.

New condition family	Question–condition answers	Logical calls
Matched Direct	1000	2200
C00 / C10 / C01	3000	6600
Contemporaneous Astra C11	500	1100
Conservative geometry	1000	2200
Age 0 / 1 / 2 (T5)	600	3000
Resolution 256 / 512 / 768	3000	6600
Lighting / material, both methods	4000	8800
Total	**13100**	**30500**

Table 20. **New inference accounting.** Qwen adds 15 conditions, 6,300 answers, and 14,700 calls; Astra adds 16 conditions, 6,800 answers, and 15,800 calls. The 13,100 answers reuse a fixed question set across conditions. Transport retries are not additional independent reasoning calls. Four frozen reference conditions are read for analysis in addition to the 31 new conditions.

G.2. Matched Controls and Geometry

Appendix A.2 defines the matched Direct and 2×2 binding protocols. Table 21 gives every configuration's Whole, Items, and changed F1 on the same denominator. Tables 22 and 23 provide paired corrections, regressions, conditional effects, and interaction contrasts. The C00-to-C11 whole differences are 2.80 points for Qwen and 0.20 for Astra. The interactions are 1.40 points with interval $[-0.80, 3.57]$ for Qwen and 2.40 with $[-0.41, 5.33]$ for Astra.

Conservative geometry discards every non-fixed current numerical center after an action, including centers that prescribed translation can legally propagate. Initial declarations, fixed-object geometry, joint declarations, and other P/L rules remain available. It measures the cost of excessive invalidation under the supported action set. Table 24 gives its complete comparison with full RAVEL.

Model	Condition	Whole (%)	Items (%)	Changed F1 (%)
Qwen3-VL-4B	Matched Direct	6.20	52.64	23.65
	C00: neither lock	21.60	78.56	62.15
	C10: program binding	23.20	79.21	63.82
	C01: evidence lock	21.40	78.86	62.64
	C11: full RAVEL	24.40	80.03	65.50
Astra	Matched Direct	78.00	96.14	95.52
	C00: neither lock	80.00	96.73	95.53
	C10: program binding	78.40	96.53	94.66
	C01: evidence lock	79.40	96.59	95.03
	C11: full RAVEL	80.20	96.59	94.57

Table 21. **Matched controls on balanced500.** Every row uses the same 500 questions, raw public initial information, legal images, state-wise schedule, and output caps. C00 retains derived state, program-measurement hints, typed findings, and deterministic answer composition. P binds program findings; L locks eligible fixed-dependency model evidence. Qwen C11 reuses the verified frozen requests; Astra C11 is the contemporaneous rerun.

Model	Δ Whole	Δ Items	Δ F1	Corrections	Regressions
Qwen3-VL-4B	$+18.20\,[13.98, 22.55]$	$+27.39\,[25.08, 29.49]$	$+41.86\,[35.25, 47.93]$	100	9
Astra	$+2.20\,[0.00, 4.57]$	$+0.45\,[-0.14, 1.07]$	$-0.94\,[-2.64, 0.55]$	21	10

Table 22. **Full RAVEL minus matched Direct.** Differences are percentage points with source-group paired 95% bootstrap intervals on balanced500. Corrections and regressions count questions that change whole correctness. Intervals use 10,000 replicates without multiplicity adjustment. Astra's Whole lower bound reaches zero.

Whole contrast (pp)	Qwen3-VL-4B [95% CI]	Astra [95% CI]
C00 − matched Direct	$+15.40\,[11.43, 19.41]$	$+2.00\,[-0.40, 4.66]$
C11 − C00	$+2.80\,[0.60, 5.11]$	$+0.20\,[-1.92, 2.15]$
C10 − C00 (P; L off)	$+1.60\,[-0.41, 3.79]$	$-1.60\,[-3.45, 0.20]$
C11 − C01 (P; L on)	$+3.00\,[0.81, 5.26]$	$+0.80\,[-1.19, 2.69]$
C01 − C00 (L; P off)	$-0.20\,[-1.24, 0.81]$	$-0.60\,[-2.21, 1.01]$
C11 − C10 (L; P on)	$+1.20\,[-0.61, 3.00]$	$+1.80\,[-0.94, 4.33]$
(C11 − C10) − (C01 − C00)	$+1.40\,[-0.80, 3.57]$	$+2.40\,[-0.41, 5.33]$

Table 23. **Conditional binding effects and interaction.** All contrasts use the same balanced500 rows and common source-group resamples. Direct-to-C00 changes several organization mechanisms jointly. The interaction is the difference between the L effect with P on and the L effect with P off.

Model	Conservative	Full	Δ Whole [95% CI]	Conserv. F1	Full F1
Qwen3-VL-4B	20.40	24.40	$+4.00\,[2.00, 6.20]$	57.45	65.50
Astra	79.20	80.20	$+1.00\,[-1.21, 3.15]$	94.71	94.57

Table 24. **Legal propagation versus conservative invalidation.** Whole and F1 are percentages; the difference is in percentage points. Conservative invalidation discards every non-fixed numerical center after an action, including centers that legal pure translation could propagate. Both models use balanced500.

G.3. Historical Findings and Complete Trajectories

The maximum age h is measured from a finding's original source state. Copying or locking a finding does not refresh that source. An over-age finding is removed from both the persistent evidence pool and the previous-state hint channel. At $h = 0$, the model still receives the current and permitted adjacent images, public initial state, action history, and numerical evaluation. The default $h = \infty$ applies only to findings that satisfy the validity and fixed-dependency rules. Table 25 reports all four ages, their paired whole-accuracy contrasts, item accuracy, state-vector accuracy, tokens, and cumulative time on the same 100 trajectories.

Table 26 separates initial-vector, state-vector, all-state, all-update, and whole-trajectory correctness. The state-vector denominator is 500, while the other trajectory counts use 100 questions. All-update correctness checks the transition-label arrays; Whole additionally requires every state value. This decomposition distinguishes an incomplete state sequence from incorrect update labels without assigning a cause to either error.

Model	h	Whole	Items	State vector	Δ Whole [95% CI]	Input tok.	Seconds
Qwen3-VL-4B	0	6.00	80.65	22.20	$-2.00\,[-5.10, 0.00]$	32,887	33.07
	1	7.00	82.69	24.80	$-1.00\,[-3.39, 0.00]$	33,422	31.79
	2	6.00	82.66	26.00	$-2.00\,[-5.10, 0.00]$	33,470	32.56
	∞	8.00	83.53	25.20	Ref.	33,540	32.54
Astra	0	62.00	96.94	81.20	$-9.00\,[-20.18, 2.22]$	36,298	50.00
	1	68.00	97.24	81.60	$-3.00\,[-6.45, 0.00]$	36,817	45.24
	2	72.00	97.42	83.60	$+1.00\,[-2.11, 4.71]$	36,854	45.60
	∞	71.00	97.29	81.60	Ref.	36,908	54.89

Table 25. **Maximum age of historical model findings.** Each row evaluates the same 100 complete T5 trajectories, 500 states, and 48 source groups. Quality metrics are percentages; differences are percentage points relative to $h = \infty$. Input tokens and cumulative seconds are per-trajectory means. Age follows the original source state and is not reset by copying. Both historical evidence channels are trimmed.

Model	Condition	Initial vector	State vectors	All states	All updates	Whole
Qwen3-VL-4B	Matched Direct	3.00	4.40	0	0	0
	RAVEL	60.00	25.20	9	12	8
Astra	Matched Direct	87.00	78.20	71	77	70
	RAVEL	91.00	81.60	71	75	71

Table 26. **Local and complete T5 correctness.** Initial vector uses 100 initial states; state-vector accuracy uses all 500 states. The last three columns count trajectories out of 100. All updates concerns the transition-label arrays; Whole requires both all state values and all updates. The first two quality columns are percentages.

G.4. Resolution and Paired Appearance Changes

Resolution tests resize each permitted image proportionally to a long edge of 256, 512, 768, or 1,024 pixels. They retain the same questions, labels, view sets, state schedule, output limits, and RAVEL code. Only RAVEL is evaluated in this intervention. Table 27 reports all quality metrics, paired whole contrasts, actual input tokens, and cumulative time.

Appearance tests restore the saved meshes, poses, and cameras and rerender the original physical states without rerunning dynamics. The lighting condition sets the world background to RGB $(1, 0.79, 0.61)$ at strength 0.25, multiplies existing light energy by 0.55, and sets its color to $(1, 0.83, 0.67)$. An additional area light has power 850 W, size 3, position $(1.8, -1.8, 3.0)$, and target $(0, 0, 0.55)$. The material condition assigns base colors deterministically by material name, with each channel in $[0.22, 0.77]$, replaces base-color connections, and sets roughness to 0.25 or 0.80. Friction, mass, geometry, transparency, and displacement are unchanged. Each condition uses one fixed alternative configuration across all 500 questions, with the original labels, cameras, and identity boxes.

Table 28 compares both methods under all three appearances. Relative to original images, RAVEL whole accuracy changes by -0.40 and -0.20 points for Qwen under lighting and material, and by -1.00 and $+1.40$ points for Astra. All four paired intervals span zero. The change in the RAVEL-minus-Direct advantage is $+0.80$ points for each Qwen perturbation and -0.80 and $+0.60$ for Astra, again with intervals spanning zero. These contrasts describe the tested fixed appearances and do not constitute equivalence tests.

Model	Long edge	Whole	Items	F1	Δ Whole [95% CI]	Input tok.	Seconds
Qwen3-VL-4B	1024	24.40	80.03	65.50	Ref.	14,018	13.59
	768	24.40	79.97	64.75	$+0.00\,[-1.80, 1.67]$	11,733	13.27
	512	23.00	79.64	63.38	$-1.40\,[-3.50, 0.61]$	10,101	12.87
	256	24.60	79.25	61.07	$+0.20\,[-1.83, 2.17]$	9,272	13.06
Astra	1024	80.20	96.59	94.57	Ref.	15,250	24.22
	768	76.40	96.11	94.49	$-3.80\,[-6.55, -1.29]$	12,510	22.27
	512	79.60	96.82	94.96	$-0.60\,[-2.75, 1.46]$	10,551	16.79
	256	78.20	96.28	93.87	$-2.00\,[-4.51, 0.42]$	9,375	16.24

Table 27. **Input resolution and realized cost.** RAVEL alone is evaluated on balanced500 at each resolution. Quality metrics are percentages; differences are percentage points relative to 1,024 pixels. Aspect ratio, views, labels, and schedule are unchanged. Tokens and cumulative seconds are per-question means.

Model	Appearance	Matched Direct	RAVEL	Δ Whole [95% CI]
Qwen3-VL-4B	Original	6.20	24.40	$+18.20\,[13.98, 22.55]$
	Lighting	5.00	24.00	$+19.00\,[14.80, 23.27]$
	Material	5.20	24.20	$+19.00\,[14.64, 23.53]$
Astra	Original	78.00	80.20	$+2.20\,[0.00, 4.57]$
	Lighting	77.80	79.20	$+1.40\,[-1.40, 4.24]$
	Material	78.80	81.60	$+2.80\,[0.39, 5.38]$

Table 28. **Paired appearance tests.** Whole accuracy (%) and RAVEL-minus-Direct differences (pp) use the same balanced500. Lighting and material each use one fixed alternative configuration with the original physical states, cameras, annotations, and identity boxes. No physical episode is rerun.

G.5. Relation Updates and Subgroup Results

Balanced500 contains 9,169 scored semantic items: 5,420 state-wise relation values, 3,249 T3/T5 update labels, 100 T1 transition labels, and 400 T4 candidate-membership decisions. T5 contributes 5,824 items, so longer trajectories have greater weight in micro-averaged Items. The 3,249 joint-update instances comprise 700 changed, 1,274 rechecked-preserved, 1,255 unrelated-preserved, and 20 unknown. Joint accuracy requires both endpoint values and the corresponding label, while changed precision, recall, and F1 concern the label alone. Tables 29 and 30 report both views, including false change alarms on preserved instances. The 20 unknown instances provide a category-level count rather than a confidence-calibration study.

Tables 31 and 32 use 100 questions per displayed task or physical category. Table 33 counts relation values only, excluding update labels and T4 membership decisions. These descriptive slices retain the same matched conditions and expose improvements and regressions within the aggregate score.

Model	Condition	Changed	Rechecked	Unrelated	Unknown
Qwen3-VL-4B	Matched Direct	2.29	31.63	19.20	0.00
	RAVEL	52.00	67.27	88.53	80.00
Astra	Matched Direct	92.86	95.29	98.33	0.00
	RAVEL	92.14	95.45	98.25	80.00

Table 29. **Matched joint relation updating (%).** Every item requires the before value, after value, and update label to be correct together. The T3/T5 denominators are 700 changed, 1,274 rechecked-preserved, 1,255 unrelated-preserved, and 20 unknown. The unknown column is a small-sample category, not a confidence-calibration metric.

Model	Condition	Changed P	Changed R	Changed F1	Rechecked FP	Unrelated FP
Qwen3-VL-4B	Matched Direct	25.37	22.14	23.65	21.04	14.58
	RAVEL	84.27	53.57	65.50	5.26	0.24
Astra	Matched Direct	98.34	92.86	95.52	0.78	0.00
	RAVEL	97.14	92.14	94.57	1.41	0.00

Table 30. **Changed detection and false change alarms (%).** The last two columns are the fractions of the corresponding preserved gold classes predicted as changed. These label-only metrics complement the endpoint-and-update criterion in Tab. 29. All rows use balanced500.

Task	Qwen Direct	Qwen RAVEL	Astra Direct	Astra RAVEL
T1	27.00	65.00	92.00	93.00
T2	4.00	7.00	65.00	70.00
T3	0.00	14.00	75.00	77.00
T4	0.00	28.00	88.00	90.00
T5	0.00	8.00	70.00	71.00

Table 31. **Matched task breakdown (Whole, %).** Each task has 100 questions and 20 per physical category. Direct denotes matched Direct throughout this table.

Physical category	Qwen Direct	Qwen RAVEL	Astra Direct	Astra RAVEL
Carrying	3.00	34.00	92.00	95.00
Collision	2.00	9.00	69.00	73.00
Gravity / support	9.00	20.00	88.00	89.00
Joint motion	10.00	41.00	59.00	57.00
Surface motion	7.00	18.00	82.00	87.00

Table 32. **Matched physical-category breakdown (Whole, %).** Each category has 100 questions with 20 per task. Astra's joint-motion point estimate decreases.

Predicate	Values	Qwen Direct	Qwen RAVEL	Astra Direct	Astra RAVEL
direction	1439	46.49	85.68	96.87	98.26
near_center	740	76.76	88.51	93.65	93.38
on_surface	2395	70.44	69.69	94.57	94.99
contact	392	61.48	66.07	96.68	96.43
inside	110	85.45	95.45	100.00	100.00
is_open	175	48.00	100.00	100.00	100.00
joint_connected	169	100.00	100.00	100.00	100.00

Table 33. **Matched relation-value accuracy by predicate (%).** Counts are state-wise relation values, including repeated states and nulls. Update labels and T4 candidate memberships are excluded. The seven counts sum to 5,420 relation values.

G.6. Realized Cost and Finding Provenance

Table 34 reports actual cost under matched calls and caps. Qwen's RAVEL-to-Direct ratios are 1.26 for input tokens, 3.99 for output tokens, and 3.45 for cumulative inference seconds. Astra's ratios are 1.29, 2.04, and 1.17, respectively. API reasoning tokens are already part of output accounting, and cumulative seconds sum calls within a question. Runtime and service conditions also affect recorded time.

Table 35 rereads findings from the original 4,000-question RAVEL runs, independently of the balanced500 follow-up inference. Each backend has 55,439 findings: the 46,439 scorable values cover fresh model findings, program findings, and fixed reuse. The unaligned 9,000 T4 branch findings are excluded from source-specific relation accuracy; the final T4 candidate answers retain their formal scores. Qwen's 917 wrong reuse instances descend from 428 unique initial evidence sources and Astra's 224 from 114. The unique-source counts distinguish distinct origins from repeated uses of one error.

A Qwen T5 support trace (R5) records an incorrect state-0 `on_surface` finding for fixed object D reused at states 1–3. An Astra trace (R1) provides another initial-error reuse example. These additional provenance cases are identified in the follow-up analysis; the seven complete image/request examples packaged with this kit remain those described in Appendix E.

Model	Condition	Calls/q.	Input tok./q.	Output tok./q.	Seconds/q.
Qwen3-VL-4B	Matched Direct	2.20	11,106	110.13	3.94
	RAVEL	2.20	14,018	439.91	13.59
Astra	Matched Direct	2.20	11,817	187.84	20.77
	RAVEL	2.20	15,250	382.89	24.22

Table 34. **Realized cost under matched schedules.** All rows use balanced500 and 2.2 logical calls per question. Token caps match within each backend, while actual token consumption differs. API output includes reasoning tokens. Seconds sum calls within each question and do not measure concurrent wall-clock completion.

Model	Source	All	Scorable	Accuracy	Errors	Unique origins
Qwen3-VL-4B	Fresh model	32227	26022	64.51	9236	N/A
	Program	17676	14881	100.00	0	N/A
	Fixed reuse	5536	5536	83.44	917	428
Astra	Fresh model	32244	26039	95.08	1281	N/A
	Program	17676	14881	100.00	0	N/A
	Fixed reuse	5519	5519	95.94	224	114

Table 35. **Finding provenance in the original 4,000-question RAVEL runs.** Accuracy is a percentage of scorable relation values. Each backend has 55,439 findings and 46,439 scorable values; the remaining 9,000 are T4 branch findings without aligned relation-level gold in the provenance record. T4 final answers remain scored. Unique origins count initial evidence sources of wrong fixed-reuse instances.

H. Extended Experimental Results

H.1. Evaluation Protocol

We evaluate all five backends on the same frozen list of 4,000 questions, without benchmark training. During development, 1,000 questions were individually examined and the remaining 3,000 were not individually analyzed; 99 of the 100 source groups were nevertheless exposed. The main results describe this cohort, with question-level exposure analyzed in Appendix D.1. For matched controls we use *balanced500*, a metadata-sampled subset containing 100 questions per task, 20 per task–physics cell, and all 100 source groups. Both evaluation sets assign equal weight to the five tasks. Paired bootstrap intervals resample source groups and describe group variation for fixed recorded runs; repeated-run decoding variation is unmeasured.

The primary metric is *whole-question accuracy*: every semantic answer item must be correct. Micro-averaged *answer-item accuracy* measures individual decisions over 77,303 items in the main cohort or 9,169 in balanced500. T4 contributes four candidate-membership decisions per question, with Whole requiring the exact valid set. For T3/T5, changed precision, recall, and F1 measure change detection, while *joint endpoint-and-update accuracy* requires both endpoint values and the update label to agree with the annotation. We report joint accuracy separately for changed, rechecked-preserved, and unrelated-preserved instances, and decompose complete T5 answers into state and update correctness. These metrics connect the benchmark's relation-level requirements to its complete answers; Appendix A.3 defines typed scoring and aggregation.

The local backends are Qwen3-VL-4B-Instruct [1], Spatial-SSRL-Qwen3VL-4B, and Molmo2-ER. The API backends, Sol and Astra, use the requested and returned service identifiers `gpt-6.1-sol` and `gpt-6-astra`, both at medium reasoning effort. In the main comparison, Direct receives the original question, all permitted images, common relation semantics, and the final schema in one call. RAVEL receives the same permitted image set plus public initial geometry and declarations, and uses the state-wise schedule in Appendix A. The harness rules are shared across backends, and both conditions use constrained structured output without answer repair or model-directed external tools. Appendix A records decoding settings, call counts, and image budgets.

H.2. Effect of the Complete Configuration

The complete RAVEL configuration improves whole-question and answer-item accuracy for every tested backend (Tab. 36). Whole accuracy rises from 3.60% to 23.40% for Qwen3-VL-4B, from 1.78% to 18.88% for Spatial-SSRL, and from 8.75% to 22.10% for Molmo2-ER. Sol improves from 62.05% to 79.00%, and Astra from 62.70% to 81.20%; both reach answer-item accuracy above 96%. All 40,000 final answers have valid format, with no missing final output or terminal execution failure, so the measured differences concern semantic correctness. These results establish the combined effect of additional declarations, state-wise inference, and executable organization in the complete configuration.

Paired outcomes locate the net gains within the same question list. For Sol, RAVEL corrects 809 questions and regresses on 131; for Astra, the counts are 851 and 111. Source-group bootstrap intervals for the gains are $[14.83, 19.13]$ and $[16.29, 20.78]$ percentage points, respectively. Because the main protocols also differ in information and calls, we next match those factors before examining the harness's organization and binding mechanisms.

H.3. Executable Organization under Matched Access

On balanced500, every control matches raw public initial information, legal image access, the state-wise schedule, and per-call output caps: 1,536 tokens for Qwen and 8,192 for Astra. Each condition uses 1,100 logical calls. Matched Direct predicts semantic decisions at the same states and has a controller that stores and assembles them, while RAVEL provides program-derived relation findings, typed model findings, and deterministic semantic projection. Qwen's full-RAVEL reference reuses the corresponding frozen responses after request matching; Astra's reference is rerun contemporaneously. All follow-up contrasts use paired source-group bootstrap intervals (Appendices A.2 and G).

RAVEL retains a larger advantage on Qwen under matched access (Tabs. 21 and 22). Whole accuracy increases from 6.20% to 24.40%, a gain of 18.20 points with a 95% interval of $[13.98, 22.55]$. Item accuracy and changed F1 increase by 27.39 and 41.86 points. Astra changes from 78.00% to 80.20%, a 2.20-point whole gain with interval $[0.00, 4.57]$, while changed F1 decreases from 95.52% to 94.57%. Thus, identical raw information and call schedules still leave a benefit from executable organization, with much greater gains for Qwen and a mixed metric outcome for Astra. Actual token use and inference time also differ under this match and are reported in Appendix I.5.

H.4. Binding Evidence and Propagating Premises

What compulsory binding adds. We cross enforced program binding, P, with the fixed-dependency model-evidence lock, L. C00 removes both constraints, C10 enables P, C01 enables L, and C11 is full RAVEL. Every cell retains derived state, program-measurement hints, typed findings, earlier evidence with provenance, and deterministic answer composition. P therefore controls compulsory adoption of available measurements, and L controls compulsory retention of eligible model findings. Tables 21 and 23 compare

	Whole-question accuracy (%)			Answer-item accuracy (%)	
Model	Direct	RAVEL	Gain (pp)	Direct	RAVEL
Qwen3-VL-4B	3.60	**23.40**	+19.80	44.66	**78.92**
Spatial-SSRL-4B	1.78	**18.88**	+17.10	37.59	**75.74**
Molmo2-ER	8.75	**22.10**	+13.35	47.45	**74.67**
Sol	62.05	**79.00**	+16.95	92.76	**96.64**
Astra	62.70	**81.20**	+18.50	92.83	**96.96**

Table 36. **Direct answering and the RAVEL harness.** All models use the same 4,000 questions and 77,303 answer items per condition. The RAVEL harness supplies additional public initial geometry and declarations, a state-wise call schedule, program binding, and evidence maintenance. Direct uses one call per question. This table measures the joint effect of these harness components under the recorded protocols. All metrics are higher-is-better.

the configurations and their conditional effects.

For Qwen, matched Direct to C00 accounts for 15.40 points of whole gain, and C00 to C11 adds 2.80 points with interval $[0.60, 5.11]$. The larger first contrast jointly changes several organization mechanisms, including computed relation hints; it does not isolate one component. Binding effects depend on the active constraints: adding P with L active improves Whole by 3.00 points, while adding L without P changes it by -0.20 points. Astra's C00 and C11 scores are 80.00% and 80.20%, with binding-effect intervals spanning zero. With P disabled, Astra follows all 2,169 program hints in both C00 and C01; Qwen overrides 26 and 25, respectively. This behavior helps explain the small marginal effect of forcing findings that Astra already adopts, while the Qwen results favor the broader executable organization bundle.

What valid geometry propagation adds. We separately replace selective propagation with conservative invalidation, discarding every non-fixed numerical center after each action, including those supported by legal pure translation. Restoring RAVEL's propagation improves Qwen Whole from 20.40% to 24.40%, a 4.00-point difference with interval $[2.00, 6.20]$, and changed F1 from 57.45% to 65.50%. Astra changes from 79.20% to 80.20%, with interval $[-1.21, 3.15]$, while changed F1 changes from 94.71% to 94.57% (Tab. 24). This intervention tests the premise-validity rule directly: retaining geometry supported by the action helps Qwen compared with discarding it and returning those judgments to the model. The smaller Astra effect again shows that the value of a harness operation depends on the backend using it.

I. Extended Analysis of Joint Relation Judgments

The gains in complete answers leave a gap between individual decisions and joint correctness. Over the main 4,000-question cohort, local models reach 74.67–78.92% answer-item accuracy with RAVEL but 18.88–23.40% Whole. A single incorrect endpoint, update, or candidate decision can invalidate the answer. We examine relation-level decisions, their composition across actions, and the evidence behind them to understand this gap. Figure 8 and Tab. 14 report the task breakdown; the following analyses identify whether a result uses the main cohort or matched balanced500.

I.1. Changes and Preservation

The benchmark tests preservation jointly with changes by requiring correct endpoints as well as an update label (Figure 9 and Tab. 15). On the main cohort, Qwen's changed F1 rises from 18.85% to 62.40%, while joint correctness on rechecked-preserved and unrelated-preserved instances rises from 16.94% and 38.50% to 62.68% and 87.98%. The gains therefore extend to retaining correct relation values after an interaction and outside its participation scope. Separate precision and recall also expose a trade-off for Molmo2-ER: changed recall decreases from 39.91% to 38.59%, while precision rises from 36.32% to 85.24% and F1 improves. A higher aggregate change score can thus coexist with more missed changes.

On matched balanced500, Qwen's improvement spans the three principal update classes (Tabs. 29 and 30). Changed, rechecked-preserved, and unrelated-preserved joint accuracy increases from 2.29%, 31.63%, and 19.20% to 52.00%, 67.27%, and 88.53%, respectively. False change alarms fall from 21.04% to 5.26% on affected-but-preserved relations and from 14.58% to 0.24% on unrelated preserved relations, although changed recall remains 53.57%. Astra's joint scores change from 92.86%, 95.29%, and 98.33% to 92.14%, 95.45%, and 98.25%. Its small Whole increase therefore accompanies nearly unchanged joint relation scores and slightly lower changed precision and recall. The matched evidence supports improved change-and-preservation judgments for Qwen, while Astra's aggregate gain has a different, smaller relation-level profile.

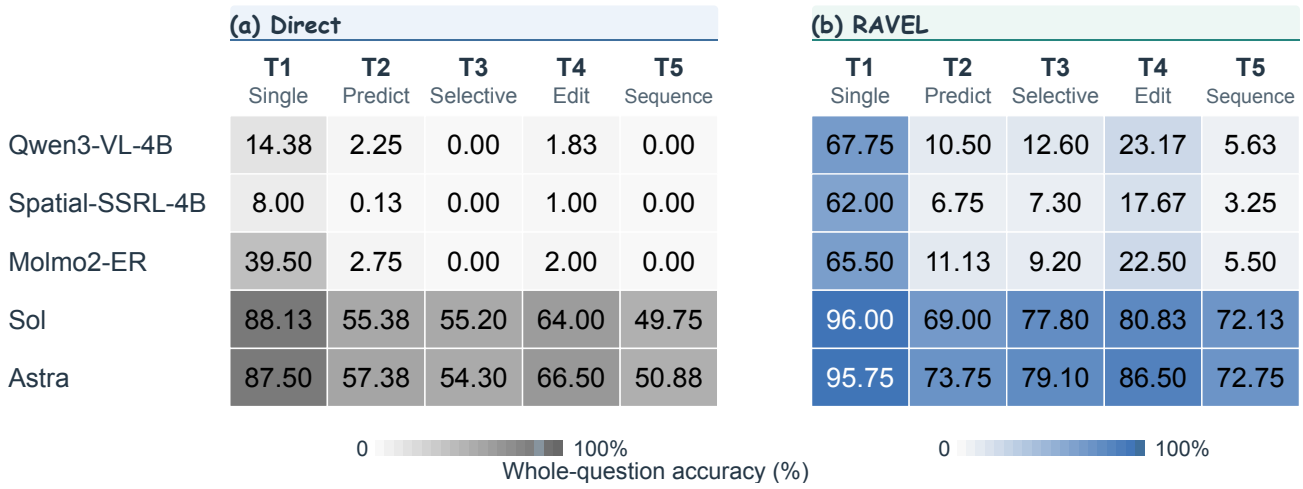


Figure 8. **Overall gains leave substantial differences among relational tasks.** Whole-question accuracy (%) for Direct and RAVEL on the same frozen questions. Both color scales cover 0–100%; cells show exact rates rounded to two decimals. T1 scores a single observed transition, T2 predicts outcomes, T3 scores multiple selective updates, T4 selects constrained edits, and T5 requires every state and update in a trajectory to be correct. Local models improve on each task while remaining below 6% on T5. This is a complete system comparison under the protocols in Appendix A.

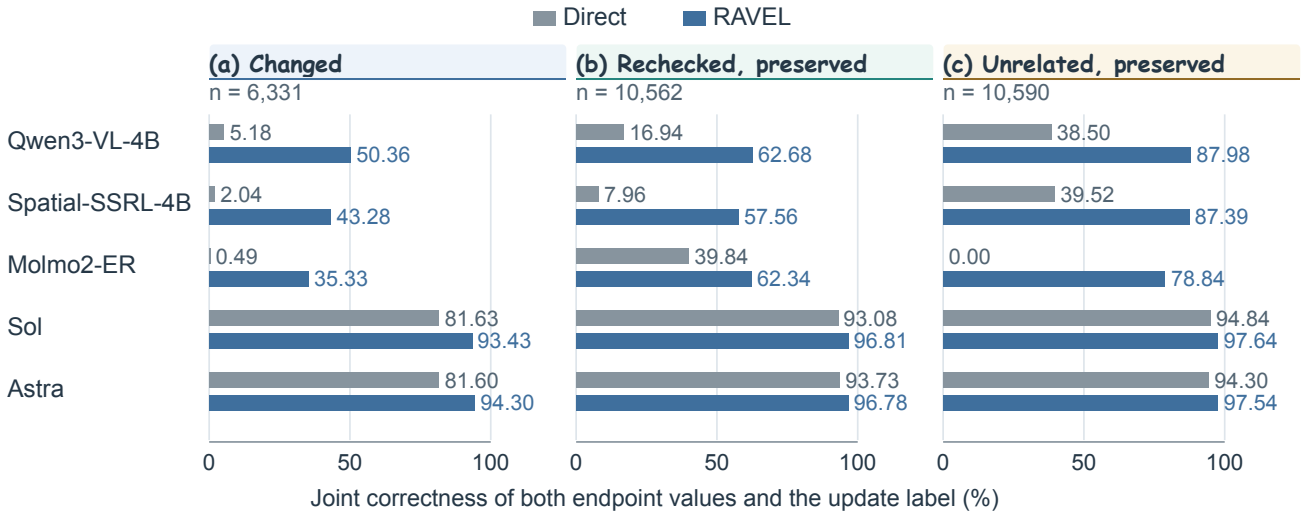


Figure 9. **Changing and preserving relations must be judged together.** Joint accuracy (%) requires both endpoint truth values and the temporal label to be correct. The three panels separate changed relations, preserved relations involving a moving participant, and preserved relations outside that participation scope. Each model pair uses the same T3/T5 instances; the 181 gold-unknown instances are excluded from these panels. Changed precision, recall, and F1, including Molmo2-ER's reduced changed recall, are retained in Tab. 15.

I.2. Complete Answers across Consecutive Actions

T5 extends joint correctness to every state and adjacent update in a trajectory. In the main cohort, RAVEL reaches 72.13% Whole for Sol and 72.75% for Astra, while the local models reach 3.25–5.63%. Astra establishes the complete initial relation vector in 728 of 800 trajectories, but only 582 have every state and update correct. The local models have no whole-correct Direct trajectories and 26–45 whole-correct RAVEL trajectories. Correctly establishing an initial scene therefore leaves substantial difficulty in producing a complete relational account of consecutive actions. Table 16 gives length and initial-correct slices; the length groups contain different questions, and the initial-correct subsets change with the method. They describe where errors occur without isolating an effect of length or initial correctness.

The 100 matched T5 trajectories separate state and update failures on the same questions (Tab. 26). Qwen's

initial-vector accuracy increases from 3% to 60%, and state-vector accuracy across 500 states increases from 4.40% to 25.20%. With RAVEL, nine trajectories have every state vector correct, twelve have every update label correct, and eight satisfy both. Astra reaches 97.29% item accuracy but 71% whole-trajectory accuracy, compared with matched Direct's 70% Whole. These decompositions show which joint requirements fail. Attributing later errors to an earlier mistaken finding requires examining the recorded evidence path, because state and update accuracy alone do not identify error propagation.

I.3. Validity and Correctness of Retained Evidence

Finding provenance exposes how computed, fresh, and retained judgments enter these answers. Each original 4,000-question Qwen or Astra RAVEL run contains 55,439 findings, of which 46,439 align with relation-value annotations; 9,000 T4 branch findings lack that alignment in the provenance record. Among scorable findings, fresh model judgments are correct in 64.51% of Qwen instances and 95.08% of Astra instances, and fixed-evidence reuse in 83.44% and 95.94% (Figure 10 and Tab. 35). The 14,881 scorable program findings per backend agree with the annotations throughout this cohort. These source-specific rates concern different queries and states; they characterize the recorded evidence rather than interchangeable inference procedures.

A finding can remain eligible for reuse while retaining an incorrect initial judgment. Qwen records 917 wrong reuse instances from 428 unique initial evidence sources, and Astra records 224 from 114. Fresh model judgments also introduce errors: 9,236 for Qwen and 1,281 for Astra. Relative to the preceding finding for the same query, fresh judgments make 3,118 correct-to-incorrect transitions for Qwen and 394 for Astra, alongside 965 and 199 incorrect-to-correct transitions. The preceding finding can itself be computed, fresh, or reused. These traces separate the persistence of an initial mistake from errors introduced by a new judgment.

How much historical evidence to retain. We intervene on retention by limiting a model finding's original-source age to $h \in \{0, 1, 2, \infty\}$ on the 100 matched T5 trajectories. Both persistent and previous-state channels discard over-age findings; copying a finding does not reset its age. Current and permitted adjacent images, initial declarations, action history, and numerical evaluation remain available. Qwen reaches 6–7% Whole with finite ages and 8% with default unlimited age, while Astra reaches 62%, 68%, 72%, and 71% (Figure 11a). Astra's $h = 0$ difference from the default is -9.00 points with interval $[-20.18, 2.22]$, and $h = 2$ has the highest point estimate. Longer retention has no monotonic advantage in these runs (Tab. 25). Together with the binding controls and provenance records, this result motivates treating continued applicability and original correctness as distinct properties of retained evidence.

Recorded relation traces. Two Astra T5 cases show how evidence becomes part of a complete answer (Figure 12; selection in Appendix E). In the translation example, Direct keeps both proximity relations true and misses their alternating values. RAVEL tracks A–C with valid numerical findings, re-evaluates B–C visually after state 0, and retains fixed furniture findings. In the tray-rotation example, Direct is correct, but RAVEL repeatedly judges B to be off the room floor after tray rotation even though the annotation remains true. These are fresh current-image support findings; the fixed-dependency lock does not apply to the moving operand. The failure therefore lies in repeated visual judgments, rather than reuse of a fixed finding.

The retained-error path appears in a Qwen support trace (R5): an incorrect state-0 `on_surface` finding for fixed D is reused at states 1–3. An Astra trace (R1) supplies another initial-error reuse trace. The cases distinguish successful premise propagation, repeated fresh misjudgment, and retained initial error. They explain actual answer paths; individual component effects are assessed by the controlled comparisons in Appendix H.3.

I.4. Unresolved Physical and Visual Judgments

Prediction and constrained selection require relation judgments without observing the outcome. On the main cohort, T2 RAVEL Whole is 6.75–11.13% for local backends and 69.00–73.75% for API backends; T4 reaches 17.67–23.17% and 80.83–86.50%, respectively. T2 requires an entire post-action relation vector, whereas T4 requires accepting every valid candidate and rejecting every invalid one under Eq. (2) of the main paper. Their different answer structures prevent interpreting the score ordering as a controlled comparison of physical prediction and constraint composition. The low local-model T2 rates show that executable organization leaves substantial difficulty in unresolved physical outcomes.

Performance also varies with the interaction and predicate being judged (Tabs. 10 and 18 to 19). On the main cohort, Astra RAVEL Whole is 95.50% on carrying, 89.63% on gravity/support, 86.63% on surface motion, 74.75% on collision, and 59.50% on joint motion. Matched balanced500 retains specific failures: Qwen reaches 7% on unseen-outcome prediction and 9% on collision, while its direction-value accuracy improves from 46.49% to 85.68% and `on_surface` changes from 70.44% to 69.69%. Astra's joint-motion Whole decreases from 59% to 57%, and its contact and proximity point estimates also decrease slightly (Tabs. 31 to 33). The benefit of organized premises thus coexists with unresolved errors in surface relations and physical consequences.

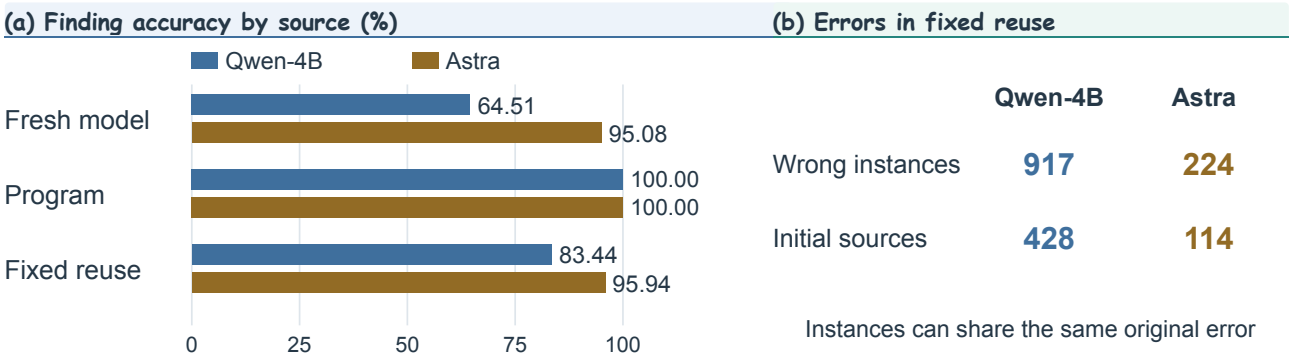


Figure 10. **Finding provenance separates fresh judgments, computed relations, and reused evidence.** The analysis rereads the original 4,000-question full-RAVEL run for each backend. (a) Relation-value accuracy among scorable findings of each source type; the program column concerns the supported rules and annotations in this cohort. (b) Wrong fixed-reuse instances and the unique initial evidence sources from which they descend. Multiple wrong instances can share one initial error. Counts, scoring coverage, and fresh-judgment errors appear in Tab. 35.

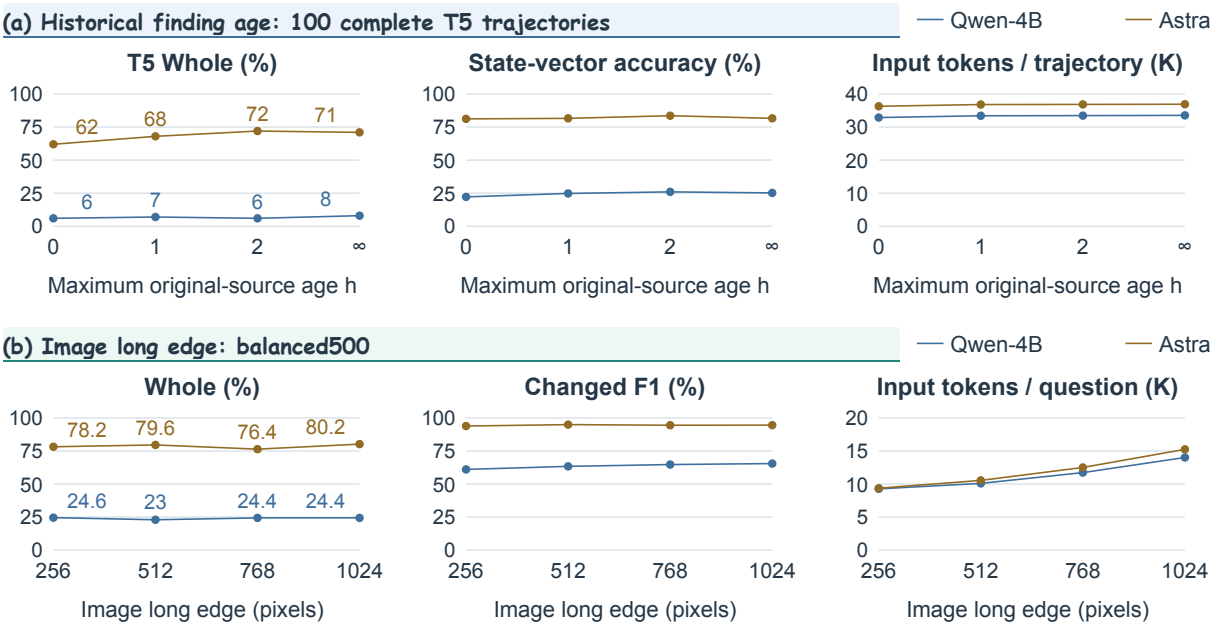


Figure 11. **Evidence retention and visual resolution affect quality and context cost differently.** (a) Maximum original-source age of model findings, evaluated on 100 complete T5 trajectories: whole accuracy, state-vector accuracy over 500 states, and mean input tokens per trajectory. (b) Image long edge, evaluated with RAVEL on balanced500: whole accuracy, changed F1, and mean input tokens per question. Token units and denominators differ between the two rows. The default settings are $h = \infty$ and 1,024 pixels. Curves connect one recorded run per condition; they do not imply monotonic effects. Full metrics, costs, and paired intervals are in Tabs. 25 and 27.

Resolution of the visual evidence. We test sensitivity to image detail by resizing each permitted image to a long edge of 256, 512, 768, or 1,024 pixels, preserving aspect ratio and the rest of RAVEL. Qwen's Whole stays between 23.00% and 24.60%, but changed F1 falls from 65.50% at 1,024 pixels to 61.07% at 256 pixels (Figure 11b). Astra's Whole varies non-monotonically: 768 pixels yields 76.40%, 3.80 points below the default, while 512 pixels yields 79.60%. Lower resolution reduces recorded input tokens, while Whole and change detection respond differently (Tab. 27). The intervention evaluates RAVEL's sensitivity to its visual input; Direct is not rerun at these resolutions.

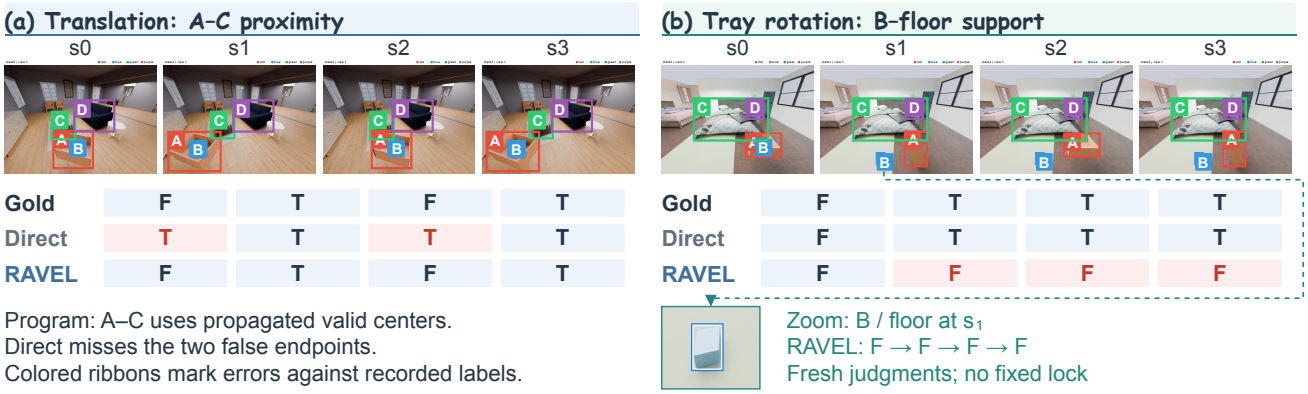


Figure 12. **Recorded correction and regression across four states.** (a) Astra RAVEL correctly alternates A–C proximity using valid geometry. (b) Its fresh visual findings miss B–floor support; Direct is correct. Red ribbons mark errors. The inset magnifies the actual state-1 image; view 1 is shown, while calls use both views. Selection is documented in Appendix E.

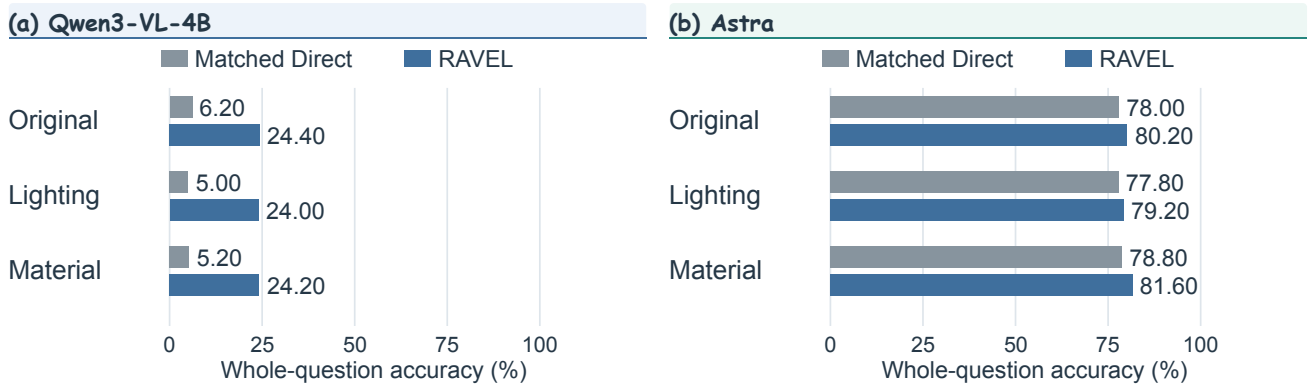


Figure 13. **Matched comparison under fixed lighting and material changes.** Whole-question accuracy on the same balanced500 episodes, with physical states, cameras, labels, and object identity boxes held fixed. Each alternative uses one deterministic rendering configuration and is evaluated with both methods and both backends. RAVEL-minus-Direct point estimates remain positive, with a larger advantage for Qwen. Table 28 reports exact differences and source-group intervals.

Appearance with physical states held fixed. We rerender balanced500 with one alternative lighting configuration and one alternative material configuration, preserving states, cameras, labels, and identity boxes, and evaluate both matched methods (Figure 13). Qwen's RAVEL Whole is 24.00% under lighting and 24.20% under material changes, with a 19.00-point advantage over matched Direct in both conditions. Astra's advantages are 1.40 and 2.80 points, with intervals $[-1.40, 4.24]$ and $[0.39, 5.38]$. The RAVEL-minus-Direct gain has a positive point estimate in both tested appearances, and paired differences from the original-image gain span zero (Tab. 28). These paired tests concern visual changes to the same physical episodes.

Both scene-source families also show gains in the main comparison. For Astra, HSSD Whole rises from 62.08% to 79.88% and Infinigen from 65.94% to 88.13% (Tab. 12). The source families differ in question allocation and overlap development; these slices describe the evaluated synthetic sources, while the paired appearance tests isolate rerendering within the same episodes. Independent source transfer remains unmeasured.

I.5. Accuracy and Inference Cost

The complete configuration's accuracy gains accompany more inference. RAVEL uses 8,995 calls per model against Direct's 4,000, and the two API backends consume about 2.75 times as many input tokens. Astra's mean input tokens per question increase from 5,626 to 15,460, and cumulative inference time from 22.18 to 33.92 seconds (Figure 14 and Tab. 17). Time sums recorded inference durations within each question before averaging, measuring cumulative inference rather than concurrent wall-clock completion.

Matched calls and output caps also leave a realized cost difference (Tab. 34). On balanced500, both methods use 2.2 calls per question, while Qwen's input tokens increase from

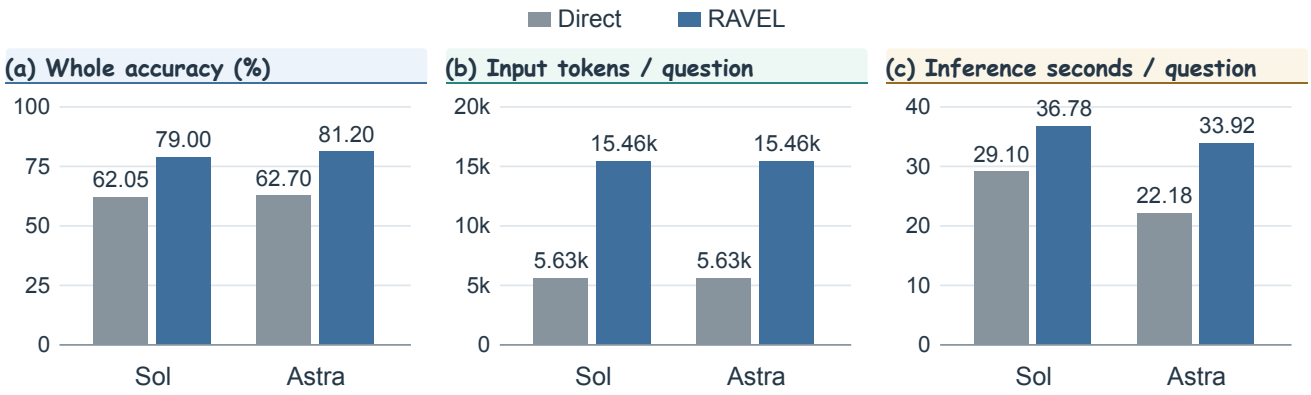


Figure 14. **Accuracy and recorded inference cost for the two API backends.** (a) Whole-question accuracy; (b) mean input tokens per question; (c) mean summed inference seconds per question. Direct uses 4,000 calls per model and RAVEL uses 8,995 under the complete system protocols. Seconds sum recorded calls rather than concurrent wall-clock completion. Tokens follow each backend's accounting. Full costs appear in Tab. 17.

11,106 to 14,018 and output tokens from 110.13 to 439.91. Its cumulative time rises from 3.94 to 13.59 seconds per question, a factor of 3.45. Astra uses 11,817 versus 15,250 input tokens, 187.84 versus 382.89 output tokens, and 20.77 versus 24.22 cumulative seconds, a time factor of 1.17. The matched experiment therefore establishes an accuracy benefit under equal information access and scheduled calls while recording the additional context and generation used to obtain it.